

Reflex Computation: Semantic Execution Blocks and Interface-Knowledge Placement

Working Draft

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Abstract

Language models can describe arithmetic procedures while still making exact execution errors. Conventional tool-use systems address this weakness by requiring the model to decide to call a calculator and serialize an invocation. We study bounded local calculator interfaces that act during generation. The original strict watcher recognizes an arithmetic suffix before an equals sign; the next iteration adds deterministic LaTeX/Unicode surface normalization and a fenced calculator block that executes only when its closing delimiter arrives. All assisted conditions share the same typed postfix micro-IR, exact-rational engine, resource bounds, and textual reinjection. After baseline-only adaptive selection, an XPU comparison used 16 arithmetic examples, 12 controls, and seven conditions. Exact-match accuracy was 43.8% for LLM-only, 62.5% for explicit tools, 75.0% for normalized reflex, and 12.5% for the original zero-shot block; normalized reflex gained five paired examples and lost none ($p = 0.0625$, unadjusted exact McNemar). We then developed block prompts on a separate split and froze a negative-first, positive-last two-shot protocol. On the same diagnostic examples, frozen blocks reached 100% valid execution and arithmetic accuracy with Qwen3-0.6B, but falsely blocked 2 of 12 controls. Qwen3-1.7B reached 93.8% execution and accuracy with zero false blocks, gaining nine and losing one pair against LLM-only ($p = 0.0215$). A Qwen3-4B six-item XPU smoke reached 100% execution with no false block, but is not a held-out estimate. Protocol-only rank-8 LoRA adapters were then trained for Qwen3-0.6B, SmoLM2-1.7B, and the approved official Gemma3-1B checkpoint. With a minimal instruction, all three adapters reached 100% block execution with zero false blocks; answers were 100%, 100%, and 93.8%, respectively. Gemma reached 100% answers when ICL was added, whereas its base ICL and normalized-runtime conditions executed only 31.2% and 25.0%. These developmental results favor a hybrid: store stable semantic selection and serialization in small adapter weights, keep exact execution and enforced result use in the runtime, and reserve context for cold-start or model-specific residual behavior. The sample remains too small for a general scaling or family claim. A frozen 120-item GSM8K transfer stress test then reversed the synthetic result: LLM-only and an un-executed block prompt both reached 10.8% accuracy, generic compute 2.5%, automatic runtime 3.3%, and LoRA block 0.8%, while an oracle bounded-calculator ledger reached 75.8%. Automatic Rescue Rate was 0/6. Thus interface syntax learned on synthetic data did not supply the natural-language decomposition needed for public transfer. This motivates an additional NL-to-ASL approach. We implement a scope-capable Pratt parser, canonical CCIR lowering, deterministic arithmetic workspace, public dataset miner, local Codex skill, and provider-neutral teacher interface. Four public sources yield 10,623 normalized records with zero scope collisions and zero teacher-facing chop failures. A revised semantic skill and answer-blind Codex pass produce 98/150 execution-verified semantic programs; two explicitly rationale-assisted repair rounds raise coverage to 150/150 (100 GSM8K and 50 TAT-QA). These are Q1 compiler-data candidates with deterministic semantic lint, not manually reviewed gold. A semantic-pattern-grouped Q1 pilot then gives 0/25 final executed answers without examples, 3/25 with five-shot ICL, 4/25 with ten-shot ICL, and 5/25, 7/25, and 7/25 for rank-8 LoRA trained on nested 25-,

50-, and 100-program subsets. LoRA-100 plus three-shot ICL reaches 10/25 (40%; 95% Wilson 23.4–59.3%). Training on 100 originals plus 900 controlled variants improves lowerability and large-number robustness but reaches only 6/25 untouched answers. A relation-diversified expansion then freezes 350 additional training programs: the resulting 450 training rows contain 397 normalized signatures and have zero signature overlap with the unchanged 25-row development and test sets. This is a learning signal and a data checkpoint, not a robust semantic compiler. Training the same rank-8 QKVO adapter on all 450 training originals then reaches 11/25 untouched answers (44%), compared with 7/25 for 100 originals; the gain supports data/semantic coverage as the first bottleneck but is concentrated in TAT-QA and does not establish a robust compiler. A clause-local adapter trained on 1,403 executed-state transitions reaches only 7/25 answers in strict predicted-state closed loop; supplying gold prior state reaches 6/25. The small, negative oracle gap and high delta validity localize the remaining failure to semantic grounding and lost global context rather than state-error propagation. Restoring the full question while retaining local-delta prediction recovers 11/25, equal to whole-program synthesis but not better. An exposure-matched OpenRouter diversity cell then replaces ten repeats of 450 programs with 4,500 unique, leakage-audited programs while retaining the same rank-8 adapter and 570 optimizer steps. Three declared training runs reach 16/25, 13/25, and 17/25 answers and 23/25, 24/25, and 22/25 executable programs, versus 11/25 and 19/25 for the frozen control. Alpha-normalized return equivalence, which ignores arbitrary internal symbol names but retains the executed expression graph, reaches 14/25, 11/25, and 14/25 versus 10/25. Strict path-sensitive semantic-state and return equivalence remain below the control. Thus semantic diversity yields a replicated autonomous answer, execution, and alpha-normalized computation gain, but not a robust canonical ontology compiler. The decisive Paper 1 comparison uses the same frozen official GSM8K identities: base Qwen3-0.6B answering directly versus the learned NL-to-ASL path followed by deterministic execution. On 250 untouched questions, direct reasoning obtains 149 correct answers (59.6%), while three U2000 adapters obtain 110, 94, and 85 (44.0%, 37.6%, and 34.0%; mean 38.5%). Thus ASL does not improve ordinary answer accuracy at this model size. On 59 pre-frozen factor-1,000 variants, however, direct reasoning degrades by 23.7 percentage points, whereas ASL changes by only -5.1 to $+3.4$ points across seeds. The resulting ASL relative-robustness gains are 18.6–27.1 points, with all three paired bootstrap intervals above zero. This supports a bounded magnitude-robustness benefit, not a general accuracy benefit. A post-hoc 2,048-token direct-reasoning sensitivity obtains 154/250 (61.6%) ordinary and 26/59 (44.1%) transformed answers; ASL’s differential robustness remains 16.9–25.4 points across seeds. Finally, a one-seed, 100-question operator pilot replaces basic arithmetic with square, cube, and exact square root. Primitive ASL plus deterministic execution obtains 100% answers versus 86% for direct reasoning while using 14.5 rather than 464 mean generated tokens. A more semantic ASL obtains only 66% because every cube case contains the same unresolved argument binding. This supports an operator-delegation signal while identifying symbol grounding as its immediate limit; the synthetic pilot is not yet an operator-complexity curve.

1 Status and Scope

This document reports an implemented next-iteration protocol, the preliminary pilot that motivated it, adaptive calibration, and one held-out comparison. The repository contains a dependency-free calculator/runtime path, deterministic dataset generation, JSONL traces and summaries, a scripted plumbing check, and an optional token-by-token Hugging Face adapter. Scripted checks are marked `empirical: false`; their expected perfect calculator behavior must not be cited as model evidence. The XPU pilot reported below is empirical but too small for model-family, scaling, or replicated performance claims. The held-out run is empirical and protocol-frozen, but it also has one checkpoint and one greedy seed; its condition ordering is evidence for replication, not a final architecture-level estimate. The public GSM8K slice is likewise a single-checkpoint developmental transfer test, not a benchmark-wide estimate. Most ASL compiler cells are single-checkpoint and

all use only 25 untouched test programs; the final data-diversity cell has three matched training runs, but its component and robustness results remain diagnostic Q1 evidence.

The initial reflex-computation phase asked:

When a language model naturally exposes a completed arithmetic expression during generation, does automatic bounded evaluation improve reliability or cost relative to model-only generation and explicit calculator calls?

The failed public-transfer result and the subsequent NL-to-ASL experiments make the mature confirmatory question more fundamental:

How much does the learned NL $\rightarrow$ ASL $\rightarrow$ **deterministic runtime** path add over the same pretrained model answering the same arithmetic questions directly, and does that contribution survive large changes in numeric magnitude?

This matched comparison is the primary Paper 1 endpoint. The earlier interface, architecture, loss, and data-scale experiments motivate and diagnose it, but do not substitute for it. We compare autonomous ASL against both concise direct answering and a stronger native-reasoning control on identical frozen questions.

The initial runtime mechanism remains calculator-only and bounded by deterministic syntax interfaces. The later ASL extension does test semantic recognition of simple arithmetic prose, but not broad-domain learned routing, persistent cross-document symbolic memory, transactional concurrency, Progressive Retrieval Attention (PRA), or native key/value access. Those mechanisms would make a positive result harder to attribute and a negative result harder to diagnose.

2 Hypotheses

H1: exact reliability. On arithmetic examples for which the strict event is correctly detected and normalized, reflex assistance increases exact-match accuracy relative to the same model without assistance.

H2: interface efficiency. Reflex or fenced-block assistance reduces model-authored invocation overhead relative to an explicit calculator condition while remaining competitive in answer quality and wall time.

H3: difficulty scaling. As operator count and operand width increase, reflex accuracy degrades more slowly than model-only and matched-prompt conditions. The relevant evidence is the condition-by-difficulty interaction or fitted scaling slope, not one easy benchmark cell.

H4: model-size leverage. Smaller models receive at least as much marginal accuracy benefit from correct reflex assistance as larger matched-family models. This is tested as a model-size-by-condition interaction, not inferred from separate unrelated checkpoints.

H5: learned-ASL contribution. On identical official GSM8K questions, autonomous ASL generation followed by validated deterministic execution improves final-answer accuracy over the same base Qwen3-0.6B answering directly. The primary baseline is native direct reasoning; concise direct answering is a supporting control. Evidence is paired by question and reported separately for every ASL training seed.

H6: magnitude robustness. When eligible source quantities are increased by a factor of 1,000 while the linguistic relations and arithmetic dependency graph are preserved, the ASL route has smaller paired accuracy degradation than direct reasoning. The primary statistic is the ASL-minus-direct difference between transformed and eligible original accuracy, with identities rather than seed-question rows resampled.

H7: ASL model-size leverage. On the identical official and factor-1,000 identities, increasing the base model from Qwen3-0.6B to Qwen3-1.7B improves the ASL-minus-direct answer contribution or its differential magnitude robustness. The estimand is the model-size difference-in-differences, $(\text{ASL}_{1.7} - \text{Direct}_{1.7}) - (\text{ASL}_{0.6} - \text{Direct}_{0.6})$, rather than the standalone ASL accuracy change. The first initialization-99,173 run is an exploratory gate; any positive interaction requires replication before a model-size claim.

H1–H4 are conditional on detection and use. Oracle detection estimates interface headroom; engine correctness alone does not demonstrate that the model will expose the syntax or use the inserted result.

3 Mechanism

3.1 Strict generation-time event

The reflex recognizes an integer arithmetic suffix completed by one equals sign, for example

The total is 37 * 48 =

and inserts the exact result before later generated text:

The total is 37 * 48 = 1776.

The event is ordinary written arithmetic, not a bespoke API token. Detection is incremental at character granularity even when a tokenizer supplies multi-character fragments. Thus, if an equals sign occurs in the middle of a fragment, the result is inserted before the remainder of that fragment is processed.

The lexical detector accepts digits, whitespace, parentheses, and +, −, *, /, //, %, and **. It requires an operator-like symbol, balanced parentheses, and a digit or closing parenthesis before the equals sign. It suppresses double-quoted candidates and rejects candidates adjacent to letters, underscores, or decimal points. Lexical detection remains separate from normalization: a candidate such as + 2 is logged and then rejected because it has no binary operation.

3.2 Typed micro-IR

The normalizer parses the candidate in expression mode and accepts only integer constants, binary operations, unary plus, and unary minus. Calls, names, attributes, indexing, containers, comparisons, booleans, floats, and all other syntax are rejected. The parsed tree is converted to a canonical postfix instruction sequence such as

PUSH 7; PUSH 5; ADD; PUSH 3; MUL.

The Python parser is used only to construct and inspect syntax. The source is never compiled or executed. A separate stack interpreter executes the micro-IR using exact rational numbers.

Table 1: Default calculator safety and complexity bounds.

Bound	Default
Expression characters	256
AST nodes / depth	64 / 16
Digits per integer literal	64
Absolute integer exponent	12
Numerator or denominator size	512 bits
Stack items	64
Engine time budget	25 ms
Recognizer suffix buffer	512 characters

3.3 Resource policy

Every event is bounded before or during evaluation. The default policy is shown in Table 1. Bounds are configuration recorded in each request, not hidden assumptions.

Division by zero, malformed IR, timeout, and resource exhaustion are typed failures. Failed results are traced but not inserted. Exponent and intermediate bit limits are checked before a large value can propagate through subsequent instructions.

3.4 Observable interrupt pipeline

Each intervention follows

DETECT $\rightarrow$ NORMALIZE $\rightarrow$ ROUTE $\rightarrow$ EXECUTE $\rightarrow$ STATE UPDATE $\rightarrow$ REINJECT.

Every transition receives a run ID, sequence number, timestamp, status, candidate/request IDs, duration where meaningful, and structured details. The append-only micro-state stores the typed request, result, creation time, detector, and engine provenance. This is enough to separate detector, parser, engine, and reinjection failures without importing the later transactional-state design.

4 Conditions

All conditions use the same arithmetic examples, checkpoint revision, decoding budget, and calculator implementation when a calculator is available.

The explicit format is deliberately unambiguous and is not counted as a reflex. Its parser and the reflex parser converge on the same arithmetic micro-IR and engine, isolating invocation style from calculator capability. A stronger native function-calling baseline should be added when the selected model/runtime exposes a stable function-calling interface; the textual baseline is the portable minimum.

5 Benchmark

5.1 Factorial arithmetic generators

The legacy deterministic generator crosses operator count with operand digit width. The separate `hard_arithmetic_v1` generator crosses at least four operator-count bands, four digit-width bands, and three structures: left chains, nested trees, and multiplication trees. It includes addition, subtraction, multiplication, safe exact-rational division, mixed operations, and long intermediates.

Table 2: Minimum experimental conditions.

Condition	Model instruction and capability	Purpose
LLM only	Return the exact answer; no calculator	Unassisted checkpoint baseline
Matched prompt	Work carefully and check operations; no calculator	Stronger prompting without external capability
Explicit tool	Model writes <code><tool:calculator>EXPR</tool></code> and receives the same engine result	Conventional model-authored invocation baseline
Reflex	Model is asked to write the expression followed by one equals sign; runtime detects any valid strict suffix	Proposed automatic condition
Normalized reflex	Reflex syntax plus deterministic LaTeX, Unicode, whitespace, and bracket normalization	Surface-robust automatic condition
Calculator block	Model writes one fenced calculator block; execution waits for the closing fence	Explicit boundary without an API schema
Oracle	Same generation instruction as reflex, but the detector accepts only the gold canonical expression	Detection/selection headroom control

Each cell uses identity-derived random seeds, and gold answers are produced by a separate exact-rational reference evaluator rather than the calculator implementation under test. The full hard grid contains

$$4 \text{ operator counts} \times 4 \text{ digit widths} \times 3 \text{ structures} = 48 \text{ cells}$$

before replicate and seed factors. Every generated item is accepted only after the independent oracle and bounded calculator agree.

This synthetic benchmark isolates scaling and component correctness; it does not establish ecological prevalence. A later extension may add held-out natural model traces, but only after freezing annotation rules for whether a generated prefix contains a valid strict event.

5.2 Non-trigger and adversarial controls

Controls include prose containing numbers, ranges, parentheses, quoted arithmetic, decimal-looking versions, variable equations, and code-like equality. Additional engine tests cover division by zero, excessive exponents and values, disallowed calls/names/indexing, and unary-only candidates. Controls are scored for false intervention and component behavior, not arithmetic answer accuracy.

5.3 Model and seed sweep

The primary manifest pins one matched Qwen3 family at 0.6B, 1.7B, and 4B parameters and uses seeds 17, 29, and 41. Repository commit hashes are stored in the configuration and copied into run manifests. The model list is an initial controlled family, not a claim of model-family generality. A second family is required before making architecture-general claims.

Generation is greedy in the initial adapter so seed variability mainly reflects future sampled configurations and experimental plumbing; if sampling is enabled, temperature, top- p , and all random states must be pinned. The correctness-first adapter recomputes the full prefix per generated token after reinjection. This is intentionally inefficient but semantically transparent. Optimized KV-cache interception is outside Paper 1 and must not be silently credited to it.

6 Metrics and Analysis

End task. Report exact rational answer accuracy with 95% confidence intervals. If a model states a final answer after receiving a calculator result, score the final claim, not merely the correct engine value. Report correction, override, and result-use rates separately.

Components. Report intended-expression exposure, candidate recognition, full-expression selection, normalization acceptance/correctness, engine success/correctness, reinjection success, result use, and later override separately. Also report false-intervention rate, intervention count, and state-item count. Oracle differences localize missed or malformed interface events.

Cost. Report prompt, model-generated, and reinjected tokens separately; model calls; engine and end-to-end wall time; and serialized trace/state growth. Do not present the correctness-first Hugging Face adapter’s full-prefix recomputation as a production latency estimate.

Scaling and inference. Report every operator-count by digit-width cell for each model, condition, and seed. Estimate condition-by-difficulty and condition-by-model-size interactions, with paired comparisons because all conditions share examples. Predeclare one primary endpoint and correct or label exploratory multiple comparisons. Preserve raw predictions and event traces so component errors can be reclassified.

7 Frozen Next-Iteration Protocol

The current interface freeze is versioned as `paper1_hard_interfaces_v2_concise`. It retains `strict_arithmetic_v1` unchanged and adds `normalized_arithmetic_v1`, `surface_normalizer_v1`, and `calculator_block_v1`. The normalizer maps allowlisted multiplication, division, Unicode-minus, bracket, and spacing aliases before delegating to the existing `ccpu.arithmetic.postfix.v1` parser. Unknown commands, prose, variables, quoted arithmetic, and quoted or inline fence examples remain inert.

The block grammar is a line-anchored Markdown-like fence named `calculator`. Its contents are collected incrementally, but no candidate is emitted until the closing fence is complete. Thus a valid intermediate line cannot execute prematurely. At close, the same surface normalizer and bounded calculator are used, and a compact typed result is inserted.

The adaptive pilot runs only the LLM-only condition over all 48 cells. Cell selection uses aggregate exact match to choose regimes near 95%, 65%, 30%, and 0%; it never examines assisted outcomes. Once those cells are selected, their definitions, prompts, grammar, normalizer, and diagnostics remain frozen. A distinct dataset seed then generates the held-out comparison across all seven conditions.

Every arithmetic prediction records expression exposure, candidate recognition, full-expression selection, normalization, execution, reinjection, result use, and later override independently. Cost records include prompt, generated, and reinjected tokens, model calls, wall and engine time, serialized trace/state bytes, and deterministic invocation-delimiter characters. Scripted smoke output is marked `empirical: false` and is plumbing evidence only.

7.1 Adaptive calibration status

Two baseline-only XPU calibrations exposed output censoring before any assisted condition was run. Both used Qwen3-0.6B, 96 arithmetic examples over 48 cells, two nominal generation seeds,

Table 3: Held-out exact-match accuracy. Intervals are 95% Wilson intervals over 16 arithmetic examples. Paired gains/losses and unadjusted exact McNemar p values compare each condition with LLM-only.

Condition	Accuracy [95% CI]	Gains-losses	Exact p
LLM only	0.438 [0.231, 0.668]	—	—
Matched prompt	0.500 [0.280, 0.720]	1-0	1.000
Explicit tool	0.625 [0.386, 0.815]	3-0	0.250
Strict reflex	0.375 [0.185, 0.614]	0-1	1.000
Normalized reflex	0.750 [0.505, 0.898]	5-0	0.0625
Calculator block	0.125 [0.035, 0.360]	2-7	0.180
Oracle	0.875 [0.640, 0.965]	7-0	0.0156

and a 48-token ceiling, for 192 generations each. Under the original prompt, all 192 generations reached the ceiling and 184 had no scorable final answer. A uniformly applied concise-answer prompt reduced cap hits to 170 and missing answers to 156, but both versions yielded only 3.125% overall exact match per seed. The concise run had 46 cells at 0%, one at 50%, and one at 100%, so it cannot supply the four intended difficulty bands.

These runs are preserved as empirical calibration failures, not treated as the hard-arithmetic comparison. Since greedy argmax decoding produced identical text for every one of the 96 cross-seed pairs, the revised calibration removes the redundant generation seed, broadens the grid downward to one-four operand digits and one-four operators, uses two independently generated examples per cell, and raises the uniform ceiling to 96 tokens.

The revised calibration completed 96 baseline-only generations with 30.2% overall exact match; 78 generations had a scorable final answer. Across 48 cells, 9 reached 100% accuracy, 11 reached 50%, and 28 reached 0%. We froze four whole cells with complete answer coverage and no cap hits: one operator/one digit with a left chain (100%), two operators/two digits with nesting (50%), one operator/three digits with a multiplication tree (50%), and one operator/four digits with a left chain (0%). The two 50% cells are explicitly coarse approximations to the overlapping moderate and transition targets.

The selection record stores source dataset and prediction hashes and states that no assisted outcome was consulted. A separately seeded held-out dataset uses four examples per selected cell, cycling ASCII, LaTeX, Unicode, and bracket surfaces, plus all 12 adversarial controls. The held-out model run remains ungenerated at this protocol-freeze point; the results below were produced only after that dataset, selection record, and interface code were committed.

8 Held-Out XPU Comparison

8.1 Setup and end-task result

The held-out run used Qwen3-0.6B at revision `c1899de`, PyTorch 2.13.0+XPU, one greedy generation seed (17), and a uniform 96-token ceiling. Four newly generated examples in each of the four frozen cells gave 16 arithmetic examples; 12 controls were crossed with the same seven conditions. All 196 prediction rows report `empirical: true` and `device: xpu`.

Normalized reflex was numerically the strongest non-oracle condition, improving by 31.25 percentage points over LLM-only and 12.5 points over explicit tools. The five paired gains and zero losses are suggestive but do not reach 0.05 in the unadjusted exact test; no multiplicity correction is applied. Strict reflex lost one paired example and gained none. The four held-out cells contain only

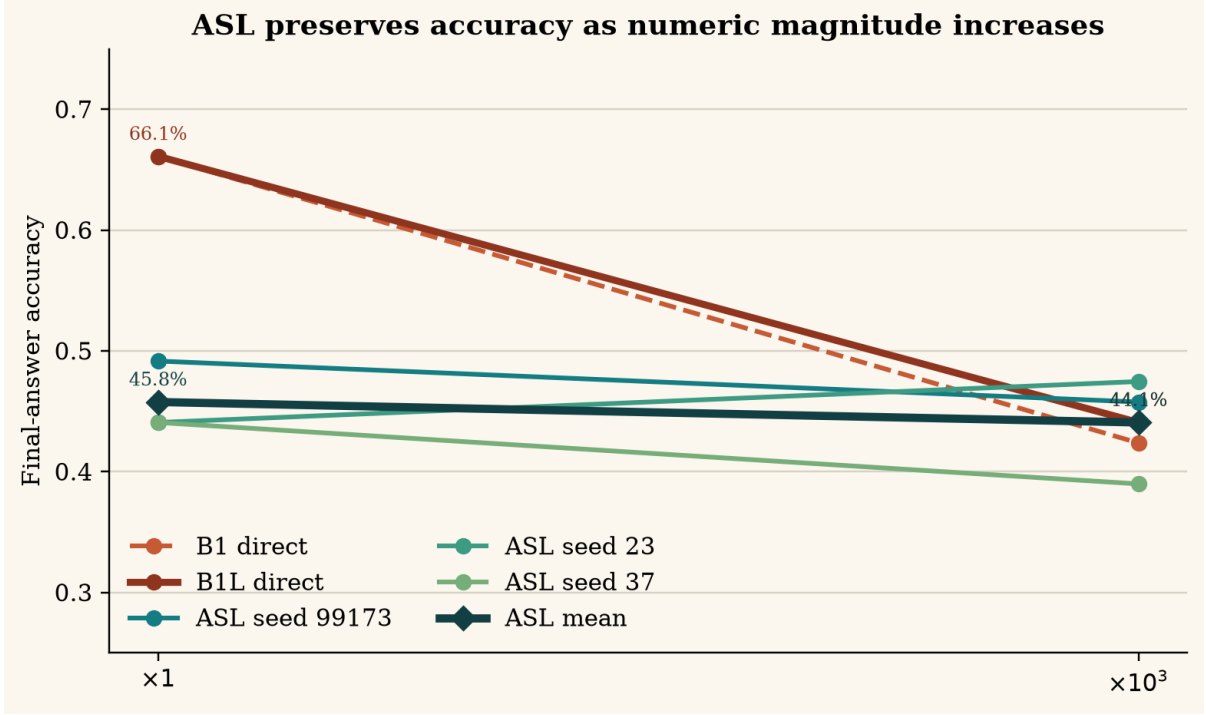


Figure 1: Matched factor-1,000 robustness on the same 59 parent questions. The direct B1/B1L controls lose 23.7/22.0 points, while the three ASL-runtime seeds remain comparatively stable. The emphasized ASL mean ties B1L at the transformed endpoint; seed-level lines retain the substantial initialization variation.

four examples each and do not show a stable monotone difficulty interaction: normalized-reflex cell accuracies were 1.00, 0.25, 1.00, and 0.75. H1 is therefore unsupported for the original strict watcher, while the normalized variant is a replication candidate rather than a confirmed claim.

8.2 Interface decomposition

Normalized reflex reached and correctly selected the full expression in 12 of 16 cases; all 12 exact results were reinjected, used, and not overridden. Strict reflex executed five candidates, but only three represented the intended full expression. Explicit tool calls were recognized in every arithmetic generation, but six arguments failed the strict parser, primarily because the model copied LaTeX or Unicode notation.

The semantic block result falsifies H3 for the tested zero-shot prompt. All 16 closing fences were recognized, but 14 blocks contained the literal placeholder `EXPRESSION` or mixed instructions/prose with arithmetic. Only two blocks normalized and executed; both engine results were exact and used. Thus the block boundary prevented premature execution, but its demonstration-style prompt elicited invalid block contents. This is a generation/normalization failure, not a calculator or closing-fence detection failure.

Cost. Mean generated tokens were 43.1 (LLM-only), 46.1 (matched), 19.2 (explicit), 47.9 (strict), 22.4 (normalized), 32.7 (block), and 15.6 (oracle). Mean wall times followed the correctness-first decoder rather than production serving: 8.0, 8.8, 3.2, 9.6, 4.0, 5.9, and 2.7 seconds, respectively. Normalized reflex used fewer delimiter characters than explicit calls, but it did not beat explicit

Table 4: Held-out assisted-interface diagnostics over arithmetic examples. Selection is correctness conditional on an accepted request; use is conditional on successful execution. All conditions had zero false interventions on 12 controls, and engine correctness was 1.0 whenever executed.

Condition	Exposure	Recognition	Selection	Normalize/execute	Use	Override
Explicit tool	0.875	1.000	1.000	0.625	1.000	0.000
Strict reflex	0.625	0.313	0.600	0.313	0.800	0.200
Normalized reflex	0.750	0.750	1.000	0.750	1.000	0.000
Calculator block	0.375	1.000	1.000	0.125	1.000	0.000
Oracle	—	1.000	1.000	1.000	0.875	0.125

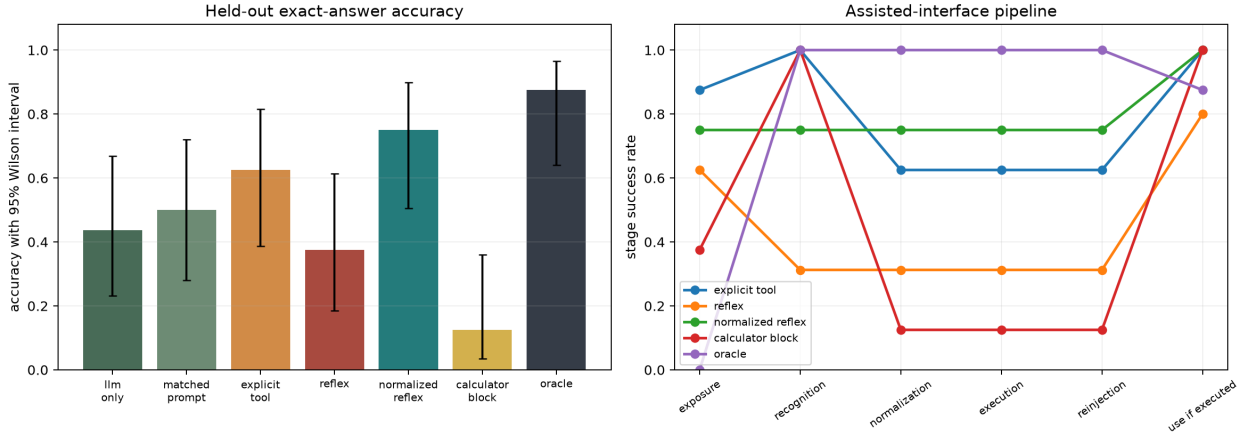


Figure 2: Held-out accuracy (left) and assisted-interface stage decomposition (right). Exposure is model-stream exposure and is not applicable to oracle priming. Use is conditional on execution, which explains high use despite low block normalization.

tools on every measured cost: explicit generated fewer tokens and was faster. H2 is therefore not supported as a quality–cost frontier claim.

8.3 Initial evidence-gate decision

At that point the decision was **no-go for Paper 2**. The original strict reflex does not improve held-out accuracy, the block interface fails zero-shot content elicitation, and the normalized-reflex advantage is unreplicated with $p = 0.0625$ over 16 pairs. A next Paper 1 replication should hold `surface_normalizer_v1` fixed, increase examples and generation seeds under sampling or add a second checkpoint family, and predeclare normalized reflex versus both LLM-only and explicit tool as primary comparisons. Block prompting or lightweight training is now eligible for a separate development split, but must not be tuned on this held-out set.

9 Few-Shot Blocks and Capability Diagnostic

9.1 Endpoint audit and prompt development

The original exact-match labels are preserved. A parallel, condition-independent endpoint extractor recognizes terminal forms such as **Response:** N without using condition identity. Under this audit the original zero-shot block’s answer accuracy changes from 12.5% to 43.8%, while normalized reflex

Table 5: Frozen ICL-G calculator blocks. The 0.6B and 1.7B rows use 16 arithmetic and 12 control examples; the 4B row is a four-arithmetic/two-control smoke only.

Model	Phase	Execute	Semantic	Answer	False controls	Use
Qwen3-0.6B	held-out diagnostic	1.000	1.000	1.000	2/12	1.000
Qwen3-1.7B	held-out diagnostic	0.938	0.938	0.938	0/12	1.000
Qwen3-4B	smoke	1.000	1.000	1.000	0/2	1.000

remains 75.0%. This is an extraction correction only: valid zero-shot block execution remains 12.5%, so the protocol conclusion is unchanged.

We tested six stronger block prompts on a separate developmental split containing four arithmetic examples and four controls. Concrete and verbatim-copy examples made arithmetic execution reliable but initially caused false blocks on all four controls; a positive/negative variant suppressed false blocks but also suppressed arithmetic blocks. A diagnostic with fully seeded fences isolated payload copying. The final order control, ICL-G, puts a negative demonstration first and a concrete positive block last. It achieved 100% semantic payload equivalence, execution, result use, and answer accuracy with no developmental false block. Exact-string payload match was 0% because the model removed redundant parentheses; the bounded parser established semantic equivalence. We froze this prompt as `paper1_calculator_block_icl_v2_order_control` before cross-model runs.

9.2 Matched-family XPU results

Table 5 compares the frozen prompt without per-model tuning. Qwen3-0.6B used the original held-out arithmetic/control set. The 1.7B checkpoint used all seven conditions over the same 28 examples, producing 196 rows. The 4B result is only the predeclared four-arithmetic/two-control smoke gate.

For 0.6B, ICL-G gained nine arithmetic examples and lost none against the preserved LLM-only run ($p = 0.0039$); however, two false blocks fail the safety part of the frozen gate. For 1.7B, it gained nine and lost one ($p = 0.0215$), while normalized reflex gained six and lost one ($p = 0.125$). The sole 1.7B block failure omitted the wrapper and emitted a bare expression. The endpoint audit leaves ICL-G at 93.8%. The unexpectedly low 1.7B oracle score (12.5% originally, 25.0% under endpoint rescoring) reflects model integration after seeded results, not calculator failure; it is retained rather than normalized away.

Family coverage and runtime. Official Llama 3.2 and Gemma checkpoints require manually accepted Hugging Face licenses. No authenticated token was available during this matched-family sweep, so four configured checkpoints were skipped and no community mirror was substituted; this capability-scaling phase is therefore Qwen-only. Approved Gemma access was supplied later for the adapter-placement phase below. Qwen3-1.7B peaked at 3.57 GB XPU memory and its full run took 3074 seconds. Qwen3-4B loaded and completed the smoke at 8.21 GB peak memory in 494 seconds. The block, accuracy/interface, token-latency, and failure plots are generated from one hashed comparison configuration and machine-readable table.

9.3 Current evidence and LoRA gate

The new result reopens the original interface gate: a frozen semantic block is reliable on the 1.7B diagnostic and clean in the 4B smoke. This is sufficient to motivate later heterogeneous-interface

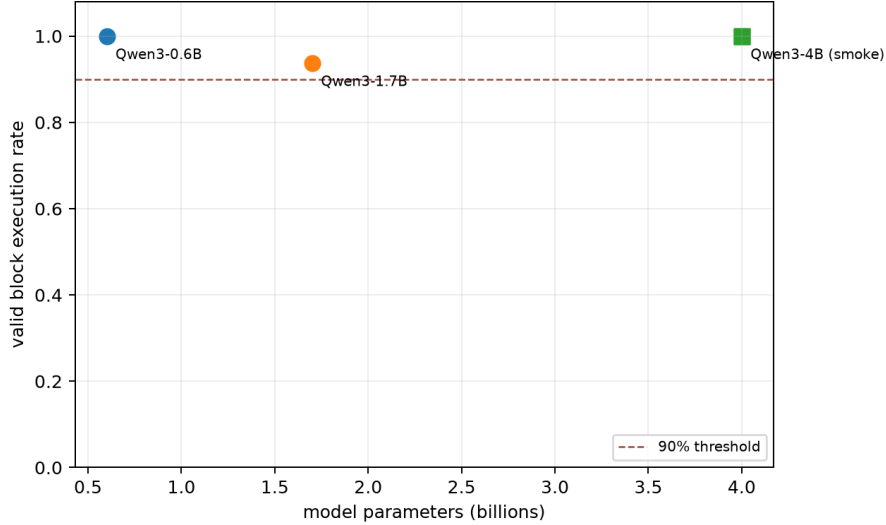


Figure 3: Valid execution under the one frozen ICL prompt. Circles are 16-example held-out diagnostics; the square is a four-example smoke and must not be read as a comparable estimate. Parameter count is descriptive, not a fitted scaling law.

work, while confirmatory Paper 1 claims remain gated on a larger untouched set and additional families. It also places the LoRA question in Regime A: can 0.6B retain its perfect arithmetic protocol while eliminating false control blocks? This asks where interface knowledge should live: transiently in context, persistently in weights, or deterministically in the runtime.

Frozen adapter protocol. The minimal protocol-only experiment uses 80 arithmetic and 80 negative-control requests; a disjoint development set has 20 of each. Arithmetic targets contain only the requested expression inside a calculator fence, never its answer. They cover three–six digit operands, nesting, mixed operations, exact division, long intermediates, and ASCII, LaTeX, Unicode, and bracket surfaces. Controls cover versions, dates, ranges, quoted arithmetic, code, identifiers, and variable-bearing equations. A mechanical audit finds no exact expression or operand overlap among training, development, and the 16-example held-out benchmark.

The XPU training gate completed PEFT forward, backward, and AdamW steps on all three models. The frozen stack adds PEFT 0.20.0 and Accelerate 1.14.0 to PyTorch 2.13.0+XPU and Transformers 4.57.6. LoRA rank is 8 over attention projections, with two epochs and assistant-target-only loss. Qwen3-0.6B is primary. The predeclared ungated second family is the official Apache-2.0 HuggingFaceTB SmolLM2-1.7B-Instruct checkpoint, not a community mirror. A later authenticated run uses the approved official Gemma3-1B checkpoint at pinned revision `dcc83ea`; Llama 3.2 access remains unavailable.

Evaluation compares base+frozen ICL (context), adapter+minimal instruction (weights), and base+normalized reflex (runtime), with base+minimal and adapter+ICL controls. It reports protocol decomposition, false blocks, answer accuracy, prompt/output tokens, training parameters/tokens/time, memory, and inference overhead.

Qwen3-0.6B adapter result. The primary adapter trained in 394.5 seconds on XPU, with 2,293,760 trainable parameters (0.383% of the model), 7,362 assistant target tokens over two epochs, and 1.48 GB peak memory. Mean training loss fell from 0.0526 to 0.000032 and development loss

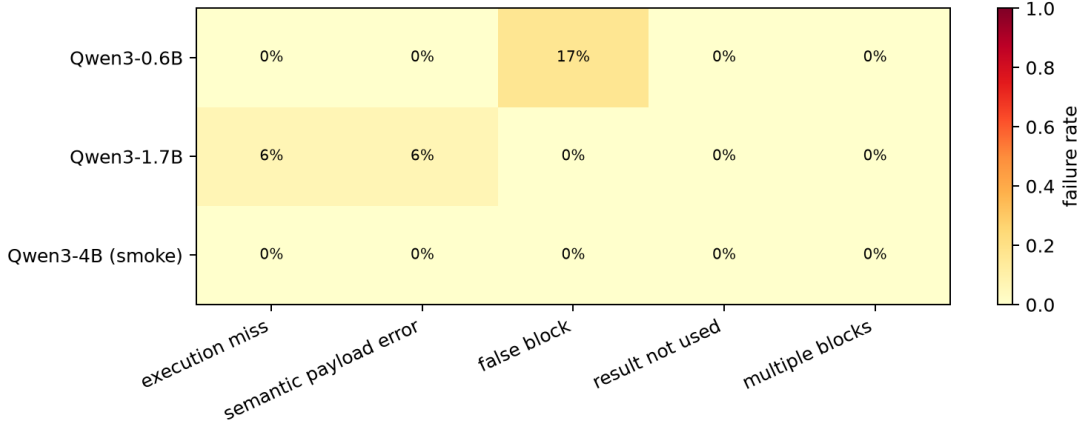


Figure 4: Block failure decomposition. The small model’s remaining error is control discrimination, whereas the 1.7B model has one arithmetic protocol miss.

Table 6: Qwen3-0.6B interface-knowledge placement on 16 arithmetic examples and 12 controls. Execute is valid arithmetic execution; false is the number of control interventions. Tokens are means over all 28 examples.

Placement/condition	Execute	Answer	False	Prompt tok.	Output tok.
Base + minimal	0.000	0.000	0/12	86.6	17.2
Context: base + ICL-G	1.000	1.000	2/12	113.8	18.9
Weights: LoRA + minimal	1.000	1.000	0/12	86.6	14.9
Weights + context: LoRA+ICL	1.000	1.000	4/12	113.8	19.2
Runtime: normalized reflex	0.750	0.750	0/12	61.0	22.4

from 0.000050 to 0.000021. The serialized adapter is 9.2 MB.

Base+minimal emitted no valid arithmetic block, so the instruction alone does not explain the adapter result. LoRA+minimal produced 16/16 semantically correct and executable payloads, used every result, answered all 16 correctly, and opened no block on 12 controls. Payload strings differed harmlessly on 14 cases, primarily by removing redundant parentheses. The adapter adds about 9.2 MB of resident model memory; measured peak inference memory increased from 1.28 to 1.42 GB. Its minimal prompt uses 27.1 fewer tokens than ICL-G and generates four fewer tokens on average.

Context and weights are not additive here. Base+ICL-G had two false blocks; LoRA+ICL-G had four, including quoted arithmetic, quoted fences, pseudocode, and quoted code. Thus the primary Qwen result favors storing the learned semantic selection/serialization contract in adapter weights with a minimal explicit contract, while retaining deterministic execution in the runtime. The normalized runtime remains safer without training but has only 75% expression exposure and execution. No result implies that LoRA learned arithmetic: every answer still comes from the unchanged bounded calculator.

Cross-family replication and placement answer. SmolLM2-1.7B used the identical data and optimization schedule. Its adapter has 3,145,728 trainable parameters (0.183%), is 12.6 MB, used 7,542 target tokens, trained in 768.0 seconds, and peaked at 3.59 GB. Training/development loss changed from 0.1287/0.0049 after epoch one to 0.0013/0.00015 after epoch two.

Table 7: Primary placement comparison. Recurring tokens sum mean prompt and generated tokens. All rates use 16 arithmetic and 12 control examples.

Model	Placement	Interface	Answer	False	Tokens
Qwen3-0.6B	Context	1.000	1.000	0.167	132.7
	Weights	1.000	1.000	0.000	101.5
	Runtime	0.750	0.750	0.000	83.5
SmolLM2-1.7B	Context	0.938	0.875	0.083	184.2
	Weights	1.000	1.000	0.000	127.6
	Runtime	0.938	0.813	0.000	96.6
Gemma3-1B	Context	0.312	0.375	0.083	131.2
	Weights	1.000	0.938	0.000	102.1
	Runtime	0.250	0.312	0.083	86.8

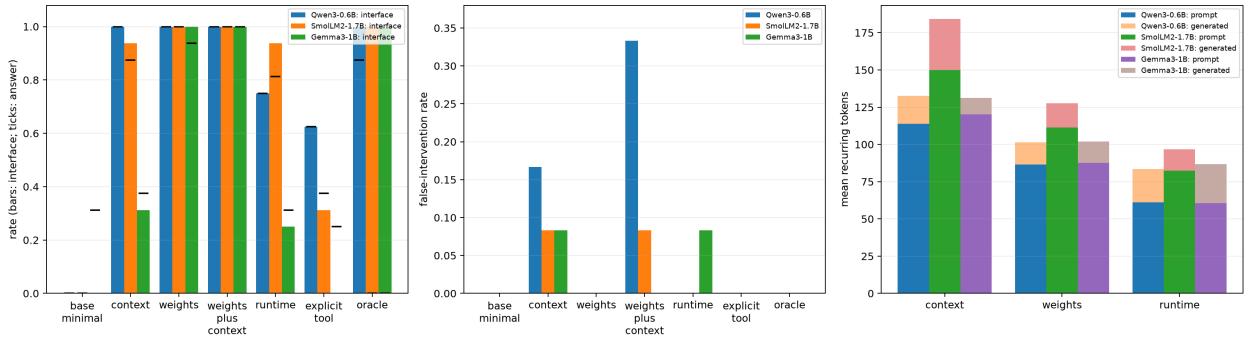


Figure 5: Interface success and answer accuracy (left), false interventions (center), and recurring prompt-plus-output tokens (right). Adapter+minimal has perfect execution and observed control safety in all three families; Gemma needs residual ICL to recover its one copied-expression error.

Base+minimal again emitted 0/16 valid blocks. Base+ICL-G executed 15/16, answered 14/16, and falsely blocked one quoted-fence control. LoRA+minimal executed and answered 16/16, used every exact result, and had 0/12 false blocks. LoRA+ICL-G also answered 16/16 but retained the quoted-fence false block and emitted multiple blocks twice. Normalized runtime executed 15/16 with no false intervention but answered 13/16 because exact result integration was imperfect.

Gemma3-1B used the same data, schedule, and rank, with its native BF16 dtype. Its 1,490,944 trainable parameters (0.149%) serialize to 6.0 MB; training used 7,362 target tokens, took 365.0 seconds, and peaked at 2.52 GB. Training/development loss changed from 0.1654/0.000094 after epoch one to 0.000069/0.000021 after epoch two. Base+minimal emitted 0/16 valid blocks. Base+ICL-G executed 5/16 and falsely blocked one control; normalized runtime executed 4/16 and also had one false intervention. LoRA+minimal executed 16/16 with 0/12 false blocks but copied one expression incorrectly, yielding 15/16 answers. LoRA+ICL-G recovered 16/16 answers and retained 0/12 false blocks. Oracle detection executed all 16 exact calculations, but Gemma used none of the reinjected results, demonstrating that runtime enforcement cannot stop at detection, execution, and reinjection.

The manual autoregressive backend originally stopped only on the tokenizer’s single EOS identifier. Gemma declares both EOS and `<end_of_turn>` in its model generation configuration. Corrected generation honors the full set {1,106}; the corrected base and adapter summaries are the only Gemma sources in this analysis. Earlier v1 diagnostic runs are retained but excluded.

The answer is therefore not one location. Stable semantic selection and block serialization are most reliable in weights under a short explicit contract in all three models. New or changing interfaces can begin in context without training, but demonstrations recur on every request and produced more false interventions; residual context repaired Gemma’s one adapter copying error but was redundant or harmful for Qwen and SmolLM. Parsing, resource bounds, exact arithmetic, reinjection, and enforced result use must remain in the runtime. This division stores policy in weights and mechanism in the runtime, with context serving development, cold start, and measured model-specific residuals. It is a developmental design result, not evidence that all interface knowledge should be fine-tuned.

10 Public GSM8K Transfer Stress Test

10.1 Frozen protocol

GSM8K contains linguistically diverse grade-school word problems with annotated multi-step arithmetic rationales [6]. We selected 120 test examples by the lowest precomputed selection key within three registered strata: 40 with at most two annotated operations, 40 with three or four, and 40 with five or more. The selection artifact redistributes only identifiers, labels, difficulty metadata, and content hashes. Questions are materialized from the pinned source at run time and rejected if their hashes change.

Qwen3-0.6B at revision `c1899de` was evaluated once with greedy seed 23011, FP16 XPU, thinking disabled, and a 160-token ceiling. Seven conditions share the same selected IDs: LLM-only, matched ICL, voluntary generic `__compute`, base calculator block, normalized automatic runtime, oracle calculator ledger, and the retained rank-8 calculator-block LoRA. Generic compute and automatic runtime use the same bounded calculator. The oracle path does not trust source-side arithmetic values: it normalizes and re-executes every supported annotated expression through that calculator before constructing the ledger. Of 425 annotated operations, 382 (89.9%) are supported; 113/120 rows have at least one supported operation and 87/120 have full trace coverage.

The primary endpoint is exact final-answer accuracy. Scorable rate records whether the model emits a parseable endpoint. Assistance rate requires at least one accepted execution, and registered-operation recall is multiset overlap with supported gold operations. The prompts were frozen after a three-item plumbing check. In particular, the generic and automatic instructions retain a concrete 7×8 syntax example; demonstration anchoring observed in the full run was not tuned away. GSM8K has no matched no-compute controls or token-level alignment, so false-activation rate and first-wrong-token prevention are undefined.

10.2 Results

The deployable interfaces do not improve over LLM-only. Voluntary generic compute accepts calls on 114/120 rows, but 168 of 225 calls repeat the demonstration expression `7*8`; its duplicate-call and malformed-row rates are 92.5% and 5.0%, respectively, and registered-operation recall is zero. Automatic runtime accepts assistance on 114/120 rows but recalls only 6.0% of supported operations. It gains two and loses eleven pairs versus LLM-only. Among six rows where voluntary compute emits no valid call, automatic runtime rescues none, so the preregistered Automatic Rescue Rate is 0/6.

The base block prompt exactly ties LLM-only at 13/120 but swaps twelve gains for twelve losses and never opens a valid block. Its row is therefore a verbose elicitation baseline, not calculator evidence. LoRA opens a valid block on 79/120 rows, but 109/120 rows contain malformed block

Table 8: Frozen Qwen3-0.6B GSM8K transfer test (120 examples). Op. recall is over calculator-supported annotated operations. Latency is the mean model wall time; interrupt conditions use correctness-first full-prefix decoding and are not production serving estimates.

Condition	Accuracy	Scorable	Assist	Op. recall	Tokens	ms
LLM only	0.108	1.000	0.000	0.000	18.7	2,776
Matched ICL	0.033	1.000	0.000	0.000	10.7	1,679
Generic compute	0.025	1.000	0.950	0.000	39.8	5,963
Base block prompt	0.108	0.383	0.000	0.000	141.1	42,396
Automatic runtime	0.033	0.992	0.950	0.060	43.9	13,148
Oracle ledger	0.758	0.983	0.942	1.000	39.9	5,907
LoRA block	0.008	0.075	0.658	0.065	55.9	20,457

activity, operation recall is 6.5%, and only nine rows emit a scorable endpoint. It answers one row correctly. The same adapter’s 16/16 synthetic held-out result therefore learned a narrow selection/serialization distribution rather than GSM8K decomposition.

The oracle ledger reaches 91/120 (75.8%, 95% Wilson interval 67.4–82.6%), with 79 paired gains and one loss versus LLM-only. This establishes large arithmetic-and-integration headroom when formalization is supplied, not a deployable routing result. Accuracy is not monotone in operation-count strata: oracle scores 60.0%, 87.5%, and 80.0%, so annotated operation count alone is not semantic difficulty.

This stress test changes the placement conclusion. Context or small adapter weights can store a stable local syntax contract, and the runtime can guarantee bounded exact execution, but none supplies the missing semantic decomposition on public word problems. Paper 1 therefore demonstrates synthetic interface feasibility and oracle public headroom, not calculator-assistance transfer.

11 ASL-Arith Dataset Bootstrap

The public transfer failure localizes the next problem before calculator execution: compile linguistic relations into an executable representation. We therefore add the pipeline

NL clause $\rightarrow$ ASL-Arith $\rightarrow$ generic AST $\rightarrow$ typed CCIR $\rightarrow$ scoped workspace $\rightarrow$ operator registry.

ASL is the compact model-facing surface, for example `month2.downloads = month1.downloads * 3`; JSON/Python dictionaries are the canonical parsed representation. The parser knows syntax and precedence but not a closed function vocabulary. Semantic lowering and the active profile registry determine types and executors. Consequently future calls, predicates, variables, rules using `:-`, queries, and `SCOPE/END` already parse without claiming Paper 2 semantics in Paper 1.

Every dataset record owns one external root scope, such as `gsm8k:train:123`. Explicit ASL scopes augment that owner; they do not replace it. Unqualified reads search the current scope and permitted ancestors, while sibling reads require qualification. Unit tests cover arithmetic precedence, state updates, explicit nested scopes, qualified sibling references, future rule/retrieval syntax, and fail-closed dangling references. Arithmetic execution also supports causal forward dependencies: a relation may be deferred until a later clause grounds its inputs, after which dependents are reevaluated.

The TAT-QA adapter now carries table and paragraph evidence into each local teacher request; without this context, metric/year names and source values cannot be grounded from the question alone. The offset-preserving chop audit finds zero record/scope collisions and zero teacher-facing failures in arithmetic-compatible question clauses. Six GSM8K, ten CLUTRR, and three

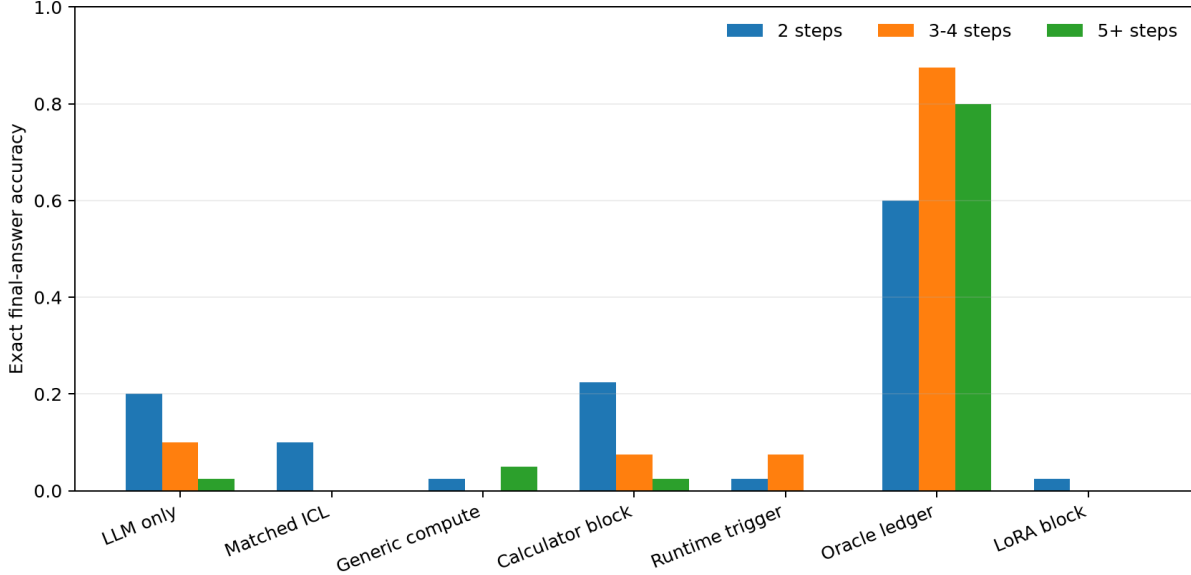


Figure 6: Exact GSM8K accuracy by registered operation-count stratum. Each bar contains 40 examples. Only the oracle ledger transfers strongly; deployable interfaces remain near zero across strata.

Table 9: Initial public DSL mining checkpoint. CLUTRR and ProofWriter are ingestion/extensibility records only and are not forced into ASL-Arith.

Dataset	Records	Arithmetic-compatible	Chopped parts
GSM8K train	7,473	7,473	52,705
TAT-QA development	1,644	706	2,477
CLUTRR test	1,146	0	7,761
ProofWriter test	360	0	7,056

ProofWriter warnings remain in non-teacher rationale or ingestion fragments and stay in the review artifact. The audit therefore opens teacher bootstrap, not model training. Deterministic stratification freezes 100 GSM8K and 50 arithmetic TAT-QA examples, producing 343 and 50 potential clause-level teacher envelopes. Each envelope carries effective scope, current state, allowed operators, source identity, source evidence when available, and the compiler-skill bundle hash. Primary requests exclude benchmark answers, gold rationales, equations, and existing programs.

Before remote teaching, a deliberately lower-quality baseline converts only dataset-provided arithmetic annotations into ASL and verifies parse, scope, execution, and final answer. It accepts 95/100 GSM8K and 50/50 TAT-QA programs. Four GSM8K annotation chains do not terminate at the benchmark answer and one has no machine-readable equation; all five are quarantined. Accepted rows are graded `Q0_DATASET_DERIVATION_EXEC_VERIFIED`: numeric execution is gold-derived, variable names are generic steps, and the programs do not prove semantic NL-to-ASL compilation.

A stricter Codex skill then requires semantic paths, measured quantities, source-versus-derived state, temporal distinctions, unresolved dependencies, and named returns. Deterministic lint rejects anonymous operation-ledger targets and literal/expression-only returns. Thirty schema-

constrained answer-blind Codex CLI batches yield 98/150 programs that pass syntax, CCIR lowering, scope, grounded execution, and hidden final-answer validation. The 52 rejects receive a labeled adjudication pass that exposes the dataset derivation as an arithmetic constraint: 45 pass after round one and seven after a second bounded repair, yielding 150/150. The final provenance split is therefore 98 primary, 45 repair-round one, and seven repair-round two. All rows remain Q1_LOCAL_CODEX_EXEC_VERIFIED; none is marked manual semantic gold.

The Click CLI can mine, audit, select, prepare answer-free Codex batches, prepare labeled repairs, validate semantic annotations, create the Q0 operation ledger, and call a configured LiteLLM teacher. Credentials remain provider-owned environment variables. The current semantic corpus used Codex CLI batches and made zero LiteLLM calls.

11.1 Frozen Q1 compiler pilot

We authorize the execution-verified Q1 corpus for a diagnostic pilot while retaining its non-manual quality label. A deterministic grouped split assigns 100 programs to training, 25 to development, and 25 to an untouched test set. Groups use renaming-invariant CCIR topology, semantic quantity family, and clause layout, so programs that differ only in entity names or constants cannot cross partitions. The frozen ledger has 92, 24, and 23 distinct pattern groups in train, development, and test, respectively, with zero pairwise overlap. Dataset counts are 66/34 GSM8K/TAT-QA in training and 17/8 in each held-out partition. The split was frozen before any model-facing run.

The experiment compares base Qwen3-0.6B, five- and ten-shot ASL prompting, rank-8 LoRA learning curves with nested 25-, 50-, and 100-program subsets, LoRA-100 plus three-shot prompting, and an augmentation adapter. The latter sees the 100 originals plus 900 deterministic, execution-verified train-only variants that rename grounded entities, remap safe numbers, and paraphrase question surfaces. Percentage parameters and calendar-year-like values are not inflated. No augmented descendant of a test or development parent enters training.

Evaluation separates the 25 untouched semantic programs from numeric remapping, very-large-number remapping, and paraphrase suites. Twenty of 25 test programs contain a safely surface-matched number eligible for each numeric suite; all 25 are eligible for paraphrase. Exact ASL is retained but is not the primary outcome: we also score parse and semantic-lint validity, entity/path grounding, source facts, operators, references and dependency edges, execution, canonical return and state equivalence under harmless path renaming, and final executed answer. With only 25 untouched programs, this pilot can reveal a learning signal and the kind of invariance learned by augmentation; it cannot establish a robust general semantic compiler.

11.2 Q1 compiler-pilot results

The 100 training programs contain 92 normalized semantic-pattern signatures: 87 occur once, three twice, one three times, and one four times. Thus the corpus is structurally diverse under the registered signature, although that signature cannot prove complete semantic diversity. Relation tags expose remaining thin classes, including only six percentage and 16 relative-quantity programs, compared with 45 ratio/fraction and 55 remaining/difference programs. Future selection therefore targets relation-class deficits rather than uniform rows.

Zero-shot Qwen3-0.6B emits natural-language summaries and no parseable ASL. Demonstrations establish the surface contract immediately but cost 9.5–16 times the base prompt tokens. LoRA-25 already reaches 5/25 answers without this prompt overhead; LoRA-50 rises to 7/25 and improves path, source-fact, operator, and state-component F1. LoRA-100 remains at 7/25 and regresses on several component scores despite a lower development loss. The curve therefore plateaus

Table 10: Untouched 25-program Q1 test. Parse and lower are validity rates; Exec. is deterministic ASL execution; Answer is final executed-answer accuracy. Exact ASL is 0 in every condition. Tokens are mean prompt tokens.

Condition	Parse	Lower	Exec.	Answer	Tokens
Base	.00	.00	.00	.00	234
Five-shot ICL	.76	.76	.64	.12	2,225
Ten-shot ICL	.88	.88	.76	.16	3,758
LoRA-25	.88	.72	.48	.20	234
LoRA-50	.92	.88	.68	.28	234
LoRA-100	.92	.84	.68	.28	234
LoRA-100 + 3-shot ICL	.88	.84	.72	.40	1,497
LoRA-B augmented	.96	.92	.64	.24	234
LoRA-500	.88	.84	.76	.44	234

Table 11: LoRA-A (100 originals) versus LoRA-B (100 originals plus 900 train-only perturbations). Numeric suites contain the 20 eligible test parents; original and paraphrase contain all 25.

Suite	LoRA-A			LoRA-B		
	Lower	Exec.	Answer	Lower	Exec.	Answer
Original	.84	.68	.28	.92	.64	.24
Numeric	.75	.55	.25	.90	.65	.25
Very large	.80	.50	.15	.90	.60	.20
Paraphrase	.80	.64	.28	.92	.64	.32

within this small test rather than supporting monotonic scaling. Three residual demonstrations complement LoRA-100, reaching 10/25; the wide Wilson interval (.234, .593) and single seed preclude a strong placement claim.

The same QKVO rank-8 adapter trained on the relation-diversified 450-row training set reaches 11/25 answers (44%; 95% Wilson interval .267-.629), with 22/25 parseable, 21/25 lowerable and type-valid, and 19/25 executable programs. This is four more answers than LoRA-100 and one more than LoRA-100 plus three-shot ICL, without demonstration tokens. Dataset breakdown is asymmetric: TAT-QA reaches 7/8 answers while GSM8K reaches 4/17. Thus added semantic originals produce a whole-program learning gain, but much of it may reflect TAT-QA specialization rather than broad GSM8K transfer.

All adapters modify 2,293,760 QKVO parameters (0.383% of the model). LoRA-25, -50, and -100 consume 29,550, 59,640, and 112,950 target tokens and finish in 427, 836, and 1,462 seconds on XPU. Their final development losses are .682, .642, and .579. LoRA-B consumes 235,470 target tokens in 3,176 seconds and reaches .535 development loss, so its comparison is not compute matched. LoRA-500 retains the same 2,293,760 trainable parameters and ten-epoch schedule, consumes 565,160 target tokens, takes 6,832 seconds, and reaches .507 development loss. It changes data coverage and total token exposure together; it is not a compute-matched ablation against LoRA-100.

Augmentation does not improve untouched semantic-program answers (6/25 versus 7/25), but it consistently improves lowerability and improves very-large-number answers from 3/20 to 4/20. Relative to its own original outputs, LoRA-B loses no correct parent under ordinary numeric remapping or paraphrase, whereas LoRA-A loses two and one, respectively. The evidence therefore supports a narrow interpretation: controlled perturbations teach surface and magnitude invariance,

not new reasoning structures. Parsing, CCIR lowering, semantic type validity, and execution are recorded separately. In particular, one parseable Boolean arithmetic output lowers mechanically but is type-invalid and cannot execute; it receives no state/execution correctness.

11.3 Relation-diversified 500-original checkpoint

We next freeze the data checkpoint that precedes any larger-adapter comparison. Selection excludes all 150 prior source IDs and greedily balances answer-free surface cues for aggregation, comparison, equal allocation, multiple entities, percentages, rates, ratios/fractions, relative quantities, remaining/difference, temporal shifts, chains, and nested dependencies. We draw a 400-row reserve (267 GSM8K and 133 TAT-QA), rather than another uniform sample. Eighty answer-blind Codex batches produce 267 programs that pass semantic lint, parse, CCIR lowering and types, grounded execution, and answer agreement. One labeled answer-visible repair round recovers 124 of 133 rejects, for 391/400 accepted programs: 267 answer-blind primary and 124 rationale-assisted repair. Nine rows remain rejected.

The post-compilation freeze recomputes each normalized semantic signature and excludes the 47 frozen development/test pattern families. Seven otherwise accepted rows collide and are quarantined. From 384 eligible programs we choose exactly 350, retaining 34 as reserve. The selected expansion contains 260 GSM8K and 90 TAT-QA programs and 314 distinct signatures; 303 signatures occur once. Combined with the original frozen 100-row training partition, the 450 training programs contain 397 signatures, with only nine signatures shared between old and new training data. Frozen development and test remain unchanged at 25 rows each, giving 500 semantic originals total and zero expansion overlap with either held-out pattern set.

The expansion campaign reports 1,320,932 primary and 642,753 repair tokens (1,963,685 total) and 1,697 plus 565 seconds of wall time under four concurrent local Codex workers. These are teacher-generation costs, not student-training tokens. Relation labels are multi-valued: among the selected 350, for example, 171 carry ratio/fraction cues, 130 relative-quantity cues, 138 percentage cues, 175 aggregation cues, and 232 chained-relation cues. Labels describe selection coverage, not independently adjudicated semantic classes. This checkpoint alone makes no student-model improvement claim. The subsequent same-rank student run, reported above, improves untouched answers from 7/25 to 11/25. We next derive 1,403 causal clause-local training transitions from the 450 programs, with only the current clause, compact evidence, scope, and prior executed state visible; the 65-transition development set retains the original 25 development parents. That incremental-state condition is the fixed input to the rank comparison below.

A separate free-route LiteLLM/OpenRouter screening campaign covers the full 7,473-item GSM8K training set and all 706 arithmetic-compatible TAT-QA development items. Both resumable queues are complete. They produce 3,885 and 515 accepted model responses, respectively (4,400 total); deterministic post-validation retains 3,489 GSM8K and 407 TAT-QA programs (3,896 total) after conversion, execution, and semantic checks. Across all generated responses, 89.0% execute to the benchmark answer and 98.7% pass semantic lint. These rates are automatic screening diagnostics, not manual semantic adjudication. The remote corpus remains provenance-separated and is excluded from every preceding training, development, and test result. Conservative recovery over the original response envelopes raises the strict pool to 6,989 unique source programs. The D1 experiment below draws a fixed, leakage-audited 4,500-program training sample from that pool without mixing remote and Codex teachers within a condition.

Table 12: Frozen 25-program whole versus incremental comparison. Pred. is the strict deployed predicted-state loop; Oracle supplies only gold prior state. Intervals are 95% Wilson intervals for final-answer accuracy.

Condition	Parse	Lower	Exec.	State	Answer	Answer interval
Whole program	.88	.84	.76	.20	11/25	[.267,.629]
Incremental pred.	.92	.92	.60	.16	7/25	[.143,.476]
Incremental oracle	.92	.88	.28	.16	6/25	[.115,.434]
Pred. + full question	.88	.88	.64	.20	11/25	[.267,.629]

11.4 Incremental executed-state compiler

The clause-local condition uses the same pinned Qwen3-0.6B revision and rank-8 QKVO placement as the whole-program adapter. Each training row contains only the current natural-language clause, compact evidence and scope, and the workspace obtained by executing verified prior ASL; answers and future clauses are hidden. The 450 training programs yield 1,403 transitions from 397 normalized semantic signatures, and the unchanged 25-program development set yields 65 transitions. Ten epochs consume 576,570 target tokens in 1,760 optimizer steps. Training takes 25,762 seconds (7.16 hours), peaks at 7.24 GiB XPU memory, and ends at train/dev loss .029/1.082. The adapter has 2,293,760 trainable parameters and its stored weight hash begins `dc7191b9`; the full provenance is machine-readable.

At test time, the deployed condition repeatedly predicts one ASL delta, validates syntax, lowering, and type safety, executes an accepted delta, and feeds the resulting workspace to the next clause. It stops at the first invalid delta and does not repair. A diagnostic oracle-state condition instead supplies the gold executed state before each clause, independently of prior student output, while leaving the current prediction untouched. Both use the same frozen 25 source IDs, model checkpoint, adapter, decoding seed, and metric definitions.

The primary predicted-state loop attempts 63 of 70 reference transitions before three fail-closed stops, accepts 60/63 (95.2%), completes all clauses in 22/25 programs, and has mean completed fraction .888. Delta parse, lower, type, and execution rates are .968, .952, .952, and .762. Yet exact operator, semantic path, source-fact, dependency, and state-delta rates are only .254, .175, .397, .175, and .365. Thus the adapter usually speaks valid ASL but often encodes the wrong local relation. Oracle state attempts all 70 transitions and accepts 65; its corresponding semantic exact rates remain .257, .300, .414, .186, and .343.

Final-answer accuracy falls from 11/25 whole-program to 7/25 incremental. Paired outcomes are seven both correct, 14 both wrong, zero whole-wrong/incremental-correct, and four whole-correct/incremental-wrong (two-sided exact binomial $p = .125$, exploratory). Oracle state reaches 6/25, a propagation gap of $-.04$: it recovers no predicted-state answer and loses one. This single-seed diagnostic therefore does not support the incremental-state-advantage hypothesis and gives no evidence that early workspace errors are the primary limitation.

The prespecified context-loss control keeps predicted-state execution but adds the untouched full original question to every local prompt. It reaches 11/25, gaining five causal-loop failures and losing one causal-loop success (two-sided exact binomial $p = .219$, exploratory). Against whole-program LoRA it has eight both correct, 11 both wrong, three gains, and three losses ($p = 1.0$). The control therefore recovers the causal chopping penalty but does not establish an incremental executed-state advantage. Parse/lower/execution are .88/.88/.64; GSM8K rises from 1/17 to 5/17 while TAT-QA remains 6/8.

The failure is sharply dataset-dependent. Predicted-state incremental retains 6/8 TAT-QA

Table 13: Matched incremental adapter-rank comparison on the frozen 25 programs. All entries except Answer are program-level rates. Full adds the untouched original question to each predicted-state prompt.

Rank	Context/state	Parse	Exec.	State	Dependency	Answer
8	Causal predicted	.92	.60	.16	.24	7/25
16	Causal predicted	1.00	.68	.12	.12	7/25
8	Causal oracle	.92	.28	.16	.20	6/25
16	Causal oracle	.92	.32	.12	.12	6/25
8	Full predicted	.88	.64	.20	.20	11/25
16	Full predicted	.84	.64	.12	.16	10/25

answers but only 1/17 GSM8K answers; oracle state retains 6/8 and 0/17. The frozen GSM8K programs average 3.65 clauses, 7.65 ASL statements, dependency depth 3.47, 2.06 entities, and 6.59 paths, versus 1.00 clause, 4.50 statements, depth 2.13, 1.50 entities, and 3.50 paths for TAT-QA. Among attempted GSM8K deltas, operator, path, and dependency exactness are .182, .182, and .109. The multi-label error audit likewise concentrates wrong paths, dependencies, operators, and returns in GSM8K. Because TAT-QA is effectively one-part in this freeze while GSM8K is genuinely sequential, this asymmetry is consistent with a local grounding and global-context bottleneck; it is not proof of a dataset-level causal explanation. The full-question control confirms that global intent matters, but its tie with whole-program synthesis leaves local semantics unresolved. The matched rank ablation below isolates adapter capacity without mistaking context removal for an architectural gain.

11.5 Adapter rank and non-data interventions

We doubled QKVO rank from 8 to 16 while holding the 1,403 training transitions, 65 development transitions, ten epochs, base-model revision, seed, decoding, and frozen test programs fixed. Rank 16 has 4,587,520 trainable parameters (0.764% of the resulting model), trains in 14,248 seconds (3.96 hours), peaks at 7.27 GiB XPU memory, and ends at train/dev loss .022/1.167. The matched rank-8 losses are .029/1.082. Thus the larger adapter fits training transitions more tightly but generalizes worse by development loss.

Doubling rank changes neither causal predicted-state nor oracle-state answers. The causal paired comparison has six both correct, 17 both wrong, one rank-8-only success, and one rank-16-only success ($p = 1.0$, exploratory). Rank 16 removes the two rank-8 program-level parse failures and raises execution from .60 to .68, but semantic state falls from .16 to .12, dependency correctness from .24 to .12, and transition operator exactness from .254 to .188. In the full-question control it reaches 10/25 rather than 11/25, with two gains and three losses ($p = 1.0$). This diagnostic does not support insufficient QKVO rank as the current bottleneck.

The state and context ablations distinguish two kinds of information. The executed workspace is mostly *extensional*: it records which paths currently have which values. The original question is *intensional*: it preserves entity roles, relation provenance, category meaning, the requested target, and why a derived value should be computed. Supplying gold extensional state changes answers from 7/25 to 6/25 at both ranks; it recovers some individual path or state deltas but no net final answers. Supplying the full narrative while keeping predicted state instead changes rank-8 from 7/25 to 11/25 and rank-16 from 7/25 to 10/25. For example, in the frozen necklace case, both causal variants treat turquoise as a literal third type/count. Full context restores the residual relation and computes $40 - 7 - 14 = 19$.

Table 14: Fine-grained whole-program LoRA-500 semantics on the frozen 25-program test. F1 is micro-averaged over the listed reference support; accuracy is correct/support for deterministically comparable relations.

Component	Score	Support	Component	Score	Support
Entity F1	.538	47	Attribute F1	.144	140
Qualifier F1	.387	67	Operator F1	.695	94
Argument order	.429	7	Relation direction	.400	30
Source-fact F1	.227	85	Dependency F1	.113	121
Aggregation/cardinality	.080	25	Temporal grounding	.300	10
RETURN target	.280	25			

This is not evidence that full context repairs workspace integration. Rank-8 full-context path exactness is .097 and rank-16 is .086; their semantic-state rates are only .20 and .12. A model can therefore recover the answer by recomputing from the question while emitting duplicated or semantically poor state. The supported conclusion is narrower: removing global intent is more damaging than accumulated state error under this representation.

The deterministic intervention analysis consequently ranks retaining the raw observed question or a model-generated intent ledger first. A gold semantic frame is unavailable at test time and is not an admissible input. Factorized path selection may use only symbols already created by accepted model output; the model must still propose unseen targets from raw text. Epoch-level development evaluation, early stopping, and regularization are also warranted by the .029/.022 training losses versus 1.082/1.167 development losses. A fixed-demonstration hybrid is supported by the earlier LoRA-100 plus three-shot gain of .12: one seed-fixed training set was used for every test item, with no per-problem retrieval. QKVO+MLP at rank 8 remains the clean adapter-location ablation after the rank plateau, and a 1–1.7B base-model replication can test base capacity. Grammar-constrained decoding is useful for fail-closed safety, but rank 16 already converts an eight-point parse gain into zero answer gain. These interventions may complement additional distinct semantic programs; none yet substitutes for an independently evaluated larger corpus.

11.6 Fine-grained semantic failure decomposition

We replayed the three frozen prediction files through a deterministic AST/CCIR diagnostic rather than generating new model output. The analysis separates surface validity and executed answers from entity, attribute, qualifier, relation, source-fact, dependency, and query grounding. Whole-program symbol alignment permits only one-to-one, lexically compatible renames with matching graph roles; incremental symbols are sticky once present in the workspace. Metrics below are therefore strict deterministic values. Twenty-six condition-program records with plausible but unresolved symbol equivalence are placed in an optional judge queue and receive no automatic credit. Generic ASL type validity does not establish dimensional compatibility, which remains explicitly not measurable under the current arithmetic type system.

The decomposition supports the prespecified broad claim but materially sharpens it. Whole-program operator mapping is stronger than strict attributes, source-fact attachment, dependencies, relation direction, aggregation, temporal qualifiers, and the returned semantic slot. In particular, only 12/30 supported directed relations and 7/25 RETURN targets are correct. Correct arithmetic can mask this weakness: all five semantically state-equivalent programs answer correctly, but 6/20 state-inequivalent programs also reach the benchmark answer. Among parse/type-valid programs, 9/21 have exact operator multisets while only 2/21 satisfy the bounded full semantic-name/graph

Table 15: Fine-grained program semantics by frozen condition. Reuse is exact reuse of an existing gold semantic path over 60 transition references and is undefined for standalone whole-program synthesis. Other columns are program-level micro-F1 except direction and RETURN, which are support-aware accuracies.

Condition	Entity	Attribute	Qualifier	Operator	Direction	Dependency	RETURN	Reuse
Whole LoRA-500	.538	.144	.387	.695	.400	.113	.280	–
Incremental pred.	.619	.201	.480	.762	.381	.125	.240	.033
Incremental oracle	.629	.435	.654	.743	.455	.139	.320	.450

equivalence test. All seven operator-perfect but path-incorrect cases nevertheless return the right answer. Thus answer agreement is not evidence of a reusable semantic workspace.

Incremental localization modestly raises entity, attribute, qualifier, and operator scores but does not improve final answers, direction, dependencies, or RETURN grounding. More importantly, the deployed predicted-state loop exactly reuses only 2/60 existing semantic references, versus 27/60 under oracle prior state. This identifies symbol continuity as a concrete stateful failure, while the oracle condition’s 6/25 answer result shows that improved naming continuity alone is insufficient. The strict attribute and dependency scores are the weakest measured whole-program components, but unresolved ontology aliases and the 25-program sample prevent claiming a universal primary cause. The safe artifacts retain dataset, semantic-family, complexity, conditional, teacher-name consistency, ID-only error examples, and input/output hashes; full records remain local under repository benchmark-text policy.

12 Matched Functor Representation Add-on

We add a representation-controlled experiment over the frozen 500-source ASL ledger. F0 is the existing assignment-style ASL. F1 uses flat, low-level functors that are deliberately isomorphic to ASL operations, for example `add(dst, lhs, rhs)`. F2 uses semantic functors such as `offset`, `remaining`, `per_unit_total`, and `percentage_ratio`; a deterministic runtime lowers these constraints to the same typed workspace operations. F1 and F2 therefore share the same function-call surface form. A gain confined to F2 is evidence for semantic decomposition rather than parentheses, shorter syntax, or a generic functor bias.

Primary annotation sees only the raw question, a condition-specific grammar, and one fixed three-example teacher prefix. Answers, rationales, ASL programs, blackboard records, intermediate values, and validator feedback are hidden. Retries are independent raw-question attempts and never receive an earlier candidate or its failure. Student training and evaluation use one fixed empty ICL prefix for every record. The runtime, not the model, canonicalizes symbols and orders F2 constraints; its bounded single-unknown solver supports both forward and inverse relations without exposing solved state to the generator.

The primary pass accepted 451/500 paired targets; independent answer-blind retries raised coverage to 483/500. Accepted programs average 6.83 calls and 427 characters for F1 versus 6.65 calls and 420 characters for F2; 82.0% of F2 programs use at least one semantic-specific functor. The longest combined student prompt and target is 1,074 tokens for F1 and 1,076 for F2, below the shared 1,152-token limit. Both conditions train the same pinned Qwen3-0.6B QKVO-r8 adapter, seed, ten-epoch schedule, and decoding protocol as F0, and are evaluated on the identical 25 test source IDs. The principal limitation is that strict answer-blind filtering leaves F1/F2 with 434 training rows rather than F0’s 450. We report this deficit instead of repairing targets from hidden

Table 16: Answer-blind functor corpus over the frozen F0 source partition. Each accepted source has independently valid F1 and F2 targets. The 17 unresolved sources remain quarantined rather than gold-repaired.

Split	Frozen sources	Paired accepted	Quarantined
Train	450	434	16
Development	25	24	1
Test	25	25	0
Total	500	483	17

Table 17: Fine-grained representation comparison on the identical frozen 25 programs with Qwen3-0.6B. Dependency and blackboard columns are per-program macro-F1; other columns are exact program rates.

Condition	Parse	Semantic structure	Dependency	Blackboard	Execute	Final
F0 ASL	.88	.08	.101	.020	.76	.44
F1 low-level	.68	.00	.082	.087	.64	.24
F2 semantic	.60	.04	.121	.081	.60	.36

gold, and treat F1–F2 as the cleaner causal contrast.

12.1 Fine-grained 0.6B representation diagnosis

We replayed the frozen predictions through a deterministic five-layer analyzer: surface validity, common semantic structure, runtime lowering, executed blackboard state, and final answer. The common reference is the query-relevant computational graph in the gold F1 program; this excludes descriptive ASL facts that are disconnected from the query. No model judge is used. Exact semantic structure requires a complete deterministic symbol alignment, relation graph, and query target. Blackboard F1 is the strict path–value fact F1 after execution, so it does not credit a numerically correct value stored under an unaligned slot.

The runtime is not the observed ceiling: all 25 gold F0, F1, and F2 programs parse, lower, execute, and return the benchmark answer. Nor is unseen vocabulary the explanation: every gold test functor occurs in F2 training. Syntax accounts for part, but not all, of the loss. Among the 15 parse-valid F2 predictions, functor-class micro-F1 is .405 (macro-F1 .377), argument-role accuracy is .388, exact argument binding is .307, relation-direction accuracy is .556, and query target accuracy is 7/15. F1 achieves 0/25 strict ASL-isomorphic computational graphs. Thus the present evidence identifies semantic class selection, binding, and symbol/query grounding rather than deterministic lowering as the principal measured deficit.

The target surfaces do not provide a simple length explanation. Gold F0, F1, and F2 average 6.64, 7.08, and 6.76 statements, respectively; F2 is slightly shorter than F1 and has similar mean arity (2.36 versus 2.35). Dataset behavior remains heterogeneous: F2 answers 5/17 GSM8K cases versus F0’s 4/17, but only 4/8 TAT-QA cases versus F0’s 7/8. With only 25 programs, counts are primary; Wilson intervals and paired exact tests are exploratory. A fully matched Qwen3-1.7B F0/F1/F2 replication is therefore used to test the prespecified model-capacity interaction rather than to reinterpret this checkpoint.

Table 18: Pretrained T5-small LoRA matrix baseline on the frozen 25-program test set. Counts are exact; the single seed is exploratory.

Cell	Best dev loss	Parse	Execute	Semantic	Answer	Repeat stop
B0 autonomous	1.494	0/25	0/25	0/25	0/25	25/25
B0 full teacher	—	0/25	0/25	0/25	0/25	25/25
B1 autonomous	1.486	0/25	0/25	0/25	0/25	25/25
B1 full teacher	—	0/25	0/25	0/25	0/25	24/25

13 Pretrained LoRA Grounding Architecture Matrix

We next test whether external symbolic training can improve autonomous NL-to-ASL compilation without training language understanding from scratch. The frozen 500-program F0 partition supplies 450 training, 25 development, and 25 untouched test programs. The first ladder uses pretrained T5-small revision `df1b051c`, freezes 85,679,360 parameters, and trains 983,040 rank-8 QKVO LoRA parameters, 1,024 source-type parameters, and 12 cross-attention gate parameters. B0 and B1 use the same separate NL/ASL encoders and separately normalized decoder cross-attention; B0 sees NL only, whereas B1 mixes full, corrupted, and absent external ASL. Checkpoint selection uses autonomous development loss only.

B0 trained for eight epochs (456 optimizer steps) in 1,606 seconds and peaked at 0.61 GB of XPU allocation. Development loss decreased monotonically from 2.653 to 1.494, but greedy test outputs collapsed into short repeated symbol cycles: mean bounded lengths were 22.6 tokens autonomously and 23.6 with the untrained full-teacher branch. Thus lower token loss alone does not establish a usable compiler, and B0 provides no evidence that teacher memory helps. Its role is the capacity-matched autonomous baseline for B1; only B1-minus-B0 on the same test identities can measure transfer from symbolic grounding.

B1 also completed eight epochs and 456 optimizer steps, taking 1,336 seconds and peaking at 0.94 GB. Its selected autonomous development loss was slightly lower than B0 (1.486 versus 1.494), but this likelihood difference did not change any behavioral endpoint: autonomous and full-teacher evaluation both remained 0/25 for parse validity, execution, semantic return, and answer. All autonomous outputs and 24/25 full-teacher outputs met the repeated-cycle stop; the remaining full-teacher output reached EOS but was still not parseable. Accordingly, the paired B1-minus-B0 autonomous grounding gain is exactly zero. The prespecified gate rejects T5 cells B2–B4: changing encoder sharing or K/V competition is not justified when external symbolic training has not produced even syntactic transfer in the simplest matched cell. We retain this negative result and next test the same idea on a pretrained causal model whose existing F0 LoRA baseline already generates executable ASL.

13.1 Qwen Causal-Model Matrix

We then ran the matched causal-model ladder with Qwen3-0.6B revision `c1899de`. Q0 is the original F0 rank-8 QKVO LoRA control. Q1 presents serialized full, partial, or absent ASL through the ordinary token context. Q2 adds separately normalized, gated side cross-attention at transformer layers 0, 7, 14, and 21. Q3 instead merges external keys and values into native attention at those layers, preserving a single softmax. Every cell uses the same frozen 450/25/25 source-ID split, initialization and generation seeds, 565,160 supervised target tokens, ten epochs, and autonomous test input; no test program, rationale, intermediate value, or record-specific demonstration is exposed to the model.

Table 19: Qwen3-0.6B grounding architecture matrix on the frozen 25-program test set. Sem. is semantic-return equivalence. The single seed is exploratory.

Cell	Trainable	Dev loss	Parse	Execute	Sem.	Answer	Time (min)
Q0 plain F0 LoRA	2.294M	0.507	22/25	19/25	7/25	11/25	113.9
Q1 context memory	2.294M	0.339	23/25	18/25	6/25	6/25	133.4
Q2 side cross-attn.	19.072M	0.486	22/25	16/25	5/25	10/25	136.3
Q3 native merged K/V	2.295M	0.332	22/25	19/25	6/25	10/25	128.2
Q3S1 capture delta	4.589M	0.343	22/25	16/25	5/25	8/25	147.1
Q3S2 separate LoRAs	4.589M	0.334	22/25	17/25	3/25	6/25	147.1
Q3S3 hybrid LoRAs	3.442M	0.349	22/25	19/25	2/25	9/25	148.1

Q2 and Q3 each recover four answers relative to Q1 (10/25 versus 6/25), so external memory placement matters when compared with serializing the same training signal into ordinary context. Neither clears the prespecified Q0 gate: the plain LoRA control remains higher on answers (11/25) and semantic return (7/25). Q2 is especially unattractive because its 16.778M memory-patch parameters raise the trainable total by $8.3\times$ over Q3 without an answer gain and reduce execution to 16/25. Q3 preserves Q0’s 19/25 execution with only 1,024 patch-specific trainable parameters, making it the parameter-efficient diagnostic architecture, not a positive winner. We therefore do not launch multi-seed confirmation for any cell from this checkpoint. The 25-program sample remains too small for a definitive null claim; subsequent experiments first increase genuine semantic coverage and retain Q0 as the frozen control.

We also complete the planned Q3 adapter-specialization controls. Q3S1 adds an ASL-specific capture delta, Q3S2 gives NL and ASL fully separate LoRAs, and Q3S3 uses modality-specific lower adapters with a shared upper adapter. None improves over Q0. Q3S3 is the strongest specialization at 9/25 answers and preserves Q0’s 19/25 execution, but reaches only 2/25 semantic returns. Q3S2 performs no better than serialized context despite doubling the LoRA budget. Thus forcing two surface languages through specialized adapters does not yield a measured autonomous semantic advantage in this configuration; lower development loss is again not predictive of answer or semantic-return improvement. These results close the single-seed specialization branch until either genuine semantic coverage or a stronger autonomous alignment objective supplies a positive gate.

13.2 Semantic-Gradient and Bottleneck Controls

We next test two inexpensive explanations for Q0’s binding errors while keeping the pinned Qwen revision, rank-8 QKVO placement, ten-epoch schedule, frozen source split, zero-shot prompt, and autonomous NL-only test fixed. M0.6 adds a pairwise objective over deterministic within-example corruptions of paths, dependencies, operators, source facts, and query targets. F4 instead predicts a JSON symbol table and expression graph that deterministically lowers to F0 ASL; its L0 condition uses ordinary token likelihood, isolating representation from semantic reweighting.

M0.6 reaches 100% pairwise training accuracy by epoch 3 and retains it at the final epoch, but loses six Q0 answers while gaining only two. Its component F1 also falls for paths (.317 versus .379), source facts (.384 versus .484), operators (.642 versus .691), dependencies/edges (.341 versus .443), and semantic state (.268 versus .434). Thus ranking the generated gold program over local AST corruptions does not transfer to autonomous decoding and degrades execution from 19/25 to 13/25. The 2.294M-parameter run takes 8,459 seconds.

M0.5 raises the loss weight on paths, operators, literals, and RETURN while down-weighting

Table 20: Frozen autonomous 25-program semantic controls. F4 Lower is successful lowering through ASL to CCIR; its JSON-to-ASL rate is 21/25. Counts are exact.

Condition	Parse	Lower	Execute	State	Return	Answer
Q0 F0 ordinary likelihood	22	21	19	5	7	11
M0.5 F0 semantic weights	21	21	18	3	6	9
M0.6 F0 hard-negative rank	22	20	13	1	2	7
F4 L0 graph bottleneck	22	20	16	0	1	2

Table 21: Frozen autonomous data-diversity comparison. All three declared D1 training runs are complete. Counts are exact on the same 25 programs.

Condition	Parse	Lower	Execute	State	Return	Answer
Q0 Codex-450 repeated	22	21	19	5	7	11
D1 OpenRouter-4,500, init 99173	25	25	23	4	5	16
D1 OpenRouter-4,500, init 23	25	25	24	3	4	13
D1 OpenRouter-4,500, init 37	24	24	22	4	6	17

other target tokens. It improves source-fact F1 (.521 versus .484), edge F1 (.470 versus .443), and operator F1 (.700 versus .691), but path F1 falls from .379 to .305 and answer accuracy falls from 11/25 to 9/25. The paired answer transition has two gains, four losses, and seven cases correct in both conditions. Its development objective reaches its unique minimum at epoch 2 (.326 weighted, .337 ordinary) and deteriorates to .512/.486 by epoch 10, despite monotonically decreasing training loss. The final-checkpoint primary run takes 7,343 seconds. We therefore retain it as a negative matched endpoint and test the independently dev-motivated two-epoch duration as a labeled replication, not as a test-selected replacement.

F4 demonstrates that a structured surface does not by itself repair grounding. Twenty-two predictions are valid JSON and 21 lower to ASL, yet only 16 execute and two obtain the answer. Both answers are GSM8K cases; TAT-QA falls from Q0’s 7/8 to 0/8. Development likelihood is best at epoch 3 (.182) and deteriorates to .223 at epoch 10 while training likelihood continues to improve; the run takes 9,655 seconds. These single-seed controls reject syntax substitution and hard-negative ranking as stand-alone solutions. They motivate separately declared semantic weighting, dev-only checkpoint selection, and a leakage-audited increase in genuine semantic training programs; no test result is used for checkpoint selection.

13.3 OpenRouter Data-Diversity Scale Control

We next hold model revision, QKVO rank, learning rate, gradient accumulation, 570 optimizer steps, zero-shot prompt, and frozen development/test identities fixed while changing training diversity. Q0 repeats 450 Codex programs for ten epochs. D1 instead partitions 4,500 distinct strict OpenRouter programs into ten disjoint 450-row logical epoch views. The selected sample contains 4,044 GSM8K and 456 TAT-QA rows spanning 4,208 normalized signatures. It has zero source-ID overlap with frozen development or test and zero overlap with the 23 frozen test signature families. No row is truncated. This is exposure matched by rows and optimizer steps, but not by target tokens: D1 uses 435,971 target tokens versus Q0’s 565,160 because the remote programs are shorter.

D1 raises final answers from 44% to 64% (95% Wilson interval .445–.798) and shifts seven identities from wrong to correct while losing two (two-sided exact McNemar $p = .180$, exploratory). The gain is concentrated in GSM8K, which rises from 4/17 to 10/17; TAT-QA changes from 7/8 to

6/8. Syntax and runtime usability improve monotonically: parse validity reaches 25/25, lowering and typing 25/25, and execution 23/25. Mean operator F1 rises from .691 to .815, source-fact F1 from .484 to .557, path F1 from .379 to .393, and semantic-state component F1 from .434 to .449; edge F1 is unchanged at .443. However, strict state equivalence falls from 5/25 to 4/25 and strict return equivalence from 7/25 to 5/25. Thus this seed provides a substantial answer/executability signal and improved soft semantic components, but not a uniform canonical-path win. To separate wrong computation from harmless symbol renaming, evaluator v2 additionally inlines local assignments and compares alpha-normalized expression graphs. It retains constants, operators, dependency structure, and external semantic families, and does not credit a constant-only program that happens to return the right answer. Under this metric, exact return equivalence rises from 10/25 to 14/25, exact state-multiset equivalence from 6/25 to 7/25, and mean state F1 from .595 to .692. Every alpha-return success in both runs also has the correct executed answer. D1 therefore improves computational semantics under internal renaming even though it aligns less often with the teacher’s canonical path vocabulary. The final epoch is the preregistered endpoint (development loss .315); epoch 9 has the lowest diagnostic development loss (.306) and is not substituted post hoc. Training takes 4,753 seconds and 4.41 GiB peak XPU memory. Independent deterministic rescoring reproduces the stored scored-prediction hash byte for byte. Seeds 23 and 37 are matched confirmatory replications.

The first replication, seed 23, independently preserves the direction of the main runtime endpoints while reducing their magnitude. It reaches 13/25 answers and 24/25 executable programs. Relative to Q0, five identities become correct and three become incorrect (exact McNemar $p = .727$, exploratory); all five formerly non-executable cases are recovered with no execution loss ($p = .0625$). Alpha-return equivalence reaches 11/25, exact alpha-state remains 6/25, and mean alpha-state F1 rises from .595 to .631. Strict state and return equivalence are 3/25 and 4/25. Training takes 4,791 seconds and ends at development loss .319; its diagnostic minimum is epoch 8 at .316. Thus seed 23 confirms improved answer, execution, and soft alpha-state behavior, but not a uniform exact-semantic gain.

The final declared replication, seed 37, reaches 17/25 answers and 22/25 executable programs. Relative to Q0, eight identities become correct and two become incorrect (exact McNemar $p = .109$, exploratory). GSM8K reaches 11/17 and TAT-QA 6/8. Alpha-return equivalence is 14/25, exact alpha-state is 7/25, and mean alpha-state F1 is .606; strict state and return equivalence are 4/25 and 6/25. Training takes 5,302 seconds, ends at development loss .311, and retains epoch 8’s .305 minimum as a diagnostic only.

Across the three initialization runs, answer counts are 16, 13, and 17 (mean 15.33/25, range 52–68%) versus Q0’s 11/25 (44%). Execution counts are 23, 24, and 22 versus 19; alpha-return counts are 14, 11, and 14 versus 10; and mean alpha-state F1 values are .692, .631, and .606 versus .595. Exact alpha-state is improved or tied at 7, 6, and 7 versus 6. Every initialization therefore preserves the direction of the answer, execution, alpha-return, and soft alpha-state effects. In contrast, strict state counts are 4, 3, and 4 versus 5 and strict return counts are 5, 4, and 6 versus 7. Because the seeds reuse the same 25 identities, we report seed means and ranges rather than pooling 75 predictions as independent tests. Local independent rescoring and a second-machine Python 3.13 rescore of seeds 23 and 37 reproduce the complete scored JSONL hashes byte for byte. The defensible conclusion is therefore a replicated gain in autonomous answers, runtime usability, and computation graphs under internal renaming—not improved agreement with the teacher’s canonical path vocabulary and not yet a broad semantic-compiler claim.

13.4 GSM8K-Only Forward Scope

The mixed D1 experiment is retained as historical evidence, but its gain is concentrated in GSM8K and its TAT-QA component introduces table/document retrieval that is outside the revised scope of this paper. All experiments after D1 therefore use simple arithmetic word problems only, beginning with GSM8K. TAT-QA is retired from new Paper 1 training, development, checkpoint selection, and confirmation; mixed checkpoints are never relabeled as GSM8K-only checkpoints.

As the first reproducible step, we freeze G1_GSM8K: 4,500 unique F0 programs selected from the final strict OpenRouter corpus. The data boundary admits only GSM8K and rejects all 480 strict TAT-QA annotations before selection. Of 6,569 strict GSM8K programs, 29 are excluded because their source identity or protected pattern intersects the historical frozen development or test data. The remaining 6,540 eligible programs span 6,375 signatures; the selected 4,500 programs span 4,500 signatures, have zero frozen source or pattern overlap, and leave 2,040 eligible programs outside the training selection. These reserve programs are not yet a registered test set. This freeze is a dataset result, not a model result: a fresh adapter must be trained before comparison with D1.

The later arithmetic-only expansion ladder adds ASDiv and overlap-audited MAWPS as candidate training diversity, reserves SVAMP primarily for adversarial testing, and reserves GSM-Plus and GSM-Symbolic for robustness and controlled generalization. MultiArith is considered only after auditing overlap with the older arithmetic-word-problem collections. The historical mixed 25-item test and its old scores remain immutable provenance, but all new paired comparisons use its frozen 17-item GSM8K projection. A deterministic split manifest removes eight TAT-QA rows from both development and test and verifies zero train/dev/test source overlap and zero train/test pattern overlap. The larger untouched confirmation set will be drawn from the official GSM8K test split and may not guide architecture or hyperparameter selection.

Before new confirmation inference, we freeze the pinned official `openai/gsm8k` test split. Its full view contains all 1,319 test identities. A 250-item diagnostic projection, selected with fixed seed 22,901, balances solution-step strata at 83 low, 84 medium, and 83 high cases. Exact normalized-question hashing finds no overlap with the U2000/E4500 training freeze. Only question text is rendered into the model prompt; benchmark rationales and answers remain scorer-only fields. Because official GSM8K supplies no gold ASL, this evaluation measures syntax, lowering, type validity, execution, and final-answer accuracy, but not teacher-program component or state equivalence. Both views are confirmation sets and may not select checkpoints or experimental settings.

After observing the first ASL seed, we add a necessary but explicitly post-hoc matched contribution protocol. On the identical 250 identities, B0 asks the pinned base Qwen3-0.6B for a concise answer with thinking disabled and a 128-token ceiling; B1 permits native reasoning with a 1,024-token ceiling. Both use greedy decoding, the same XPU FP16 backend and chat template, and a frozen condition-independent numeric endpoint scorer. Their prompts expose only the question and never ASL examples, benchmark rationales, answers, state, or intermediate values. This stronger B1 control prevents an apparent ASL gain from being manufactured by an artificially answer-constrained baseline. The primary Paper 1 estimand is the paired answer difference between these direct conditions and autonomous ASL generation followed by deterministic execution, not a comparison with the earlier nonidentical 120-item transfer test. The scorer selects the latest accepted endpoint among an explicit final-answer form, a boxed number, or a terminal number, then compares exactly as a rational after removing comma and dollar separators. Its policy identifier is stored in every direct prediction and enforced during shard merging.

We also freeze an exploratory paired large-number view before any transformed inference. A factor-1,000 rewrite changes registered digit source quantities and propagates them through every hidden benchmark arithmetic equation. Acceptance requires source-trace verification, com-

Table 22: Frozen official GSM8K confirmation for U2000/E4500. Counts are out of 250 question-only examples; answers and benchmark rationales are scorer-only.

Initialization	Parse	Lower	Type	Execute	Answer
99173	247	247	245	212	110
23	247	242	241	222	94
37	243	240	237	208	85

plete transformed-source grounding, unchanged operator signatures, successful exact replay, and an integer terminal answer. Conservative filters reject percentages, ages/calendar values, clocks, bounded time ratios, ordinals, lexicalized quantities, hyphenated numeric unit modifiers, value-role collisions, and incomplete traces. This retains 59 of the 250 parents: 26 low-, 22 medium-, and 11 high-step cases. Transformed answers span 38 four-to-six-digit, 14 seven-to-nine-digit, and seven ten-or-more-digit values. Only the transformed question is model-visible. The principal robustness statistic is the difference between each condition’s paired original-to-large accuracy change; this suite is exploratory because its design postdates the first original-set ASL result.

13.5 Official GSM8K Confirmation

All three U2000/E4500 adapters have now completed the frozen 250-question confirmation view (Table 22). Final-answer accuracy is 44.0%, 37.6%, and 34.0% (mean 38.5%, range 34.0–44.0%). Parse validity remains high at 97.2–98.8%, whereas execution is lower at 83.2–88.8%. The initialization spread is material: only 54 questions are correct under all three adapters, 113 are wrong under all three, and 83 change outcome across initializations. We therefore report seed results separately and do not pool the 750 seed-question predictions as independent observations.

As a larger single-initialization confirmation, initialization 37 obtains 509/1,319 answers (38.6%) on the complete official test split, with 1,306/1,319 parse-valid and 1,123/1,319 executable programs. This full-set result was not used for checkpoint, prompt, or hyperparameter selection and is not pooled with the three-seed 250-item analysis.

The matched direct-reasoning control obtains 149/250 answers (59.6%) on the ordinary view. Each ASL adapter is lower by 15.6, 22.0, and 25.6 percentage points, respectively; paired identity-bootstrap 95% intervals are $[-23.2, -8.4]$, $[-30.0, -14.0]$, and $[-32.8, -18.4]$ points. The corresponding exact McNemar tests give $p = 9.34 \times 10^{-5}$, 2.86×10^{-7} , and 3.99×10^{-10} . The experiment therefore rejects an ordinary-answer advantage for this 0.6B ASL route.

The factor-1,000 comparison gives a different and more specific result (Table 23). On the same 59 eligible parents, direct reasoning falls from 39/59 to 25/59 (−23.7 points; paired bootstrap 95% interval $[-37.3, -10.2]$; exact McNemar $p = .0043$). The three ASL adapters change by only −3.4, +3.4, and −5.1 points. Relative to direct reasoning, their difference-in-differences gains are 20.3, 27.1, and 18.6 points; all three paired bootstrap intervals exclude zero. The registered 1,024-token B1 ceiling was reached by 58/250 ordinary and 18/59 large generations. The post-hoc 2,048-token B1L sensitivity improves ordinary accuracy from 149/250 to 154/250 and transformed accuracy from 25/59 to 26/59. On the 59 paired parents B1L declines from 39/59 to 26/59 (−22.0 points), so the three ASL differential robustness gains remain 18.6, 25.4, and 16.9 points; their paired bootstrap intervals remain above zero.

To measure whether this effect changes smoothly with numeric scale, we freeze a second exploratory view over the common intersection of parents that passes the same conservative transformer at every factor: $\times 1$, $\times 10^2$, $\times 10^3$, $\times 10^4$, and $\times 10^6$. This retains 55 fixed parents (10 high-, 20

Table 23: Paired factor-1,000 robustness. Original accuracy is restricted to the same 59 eligible parent identities. Δ is large minus original; robustness gain is the ASL Δ minus the direct-reasoning Δ .

Condition	Original	Large	Δ (pp)	Gain (pp)
Direct reasoning	39/59	25/59	-23.7	—
Direct reasoning long	39/59	26/59	-22.0	—
ASL init 99173	29/59	27/59	-3.4	+20.3
ASL init 23	26/59	28/59	+3.4	+27.1
ASL init 37	26/59	23/59	-5.1	+18.6

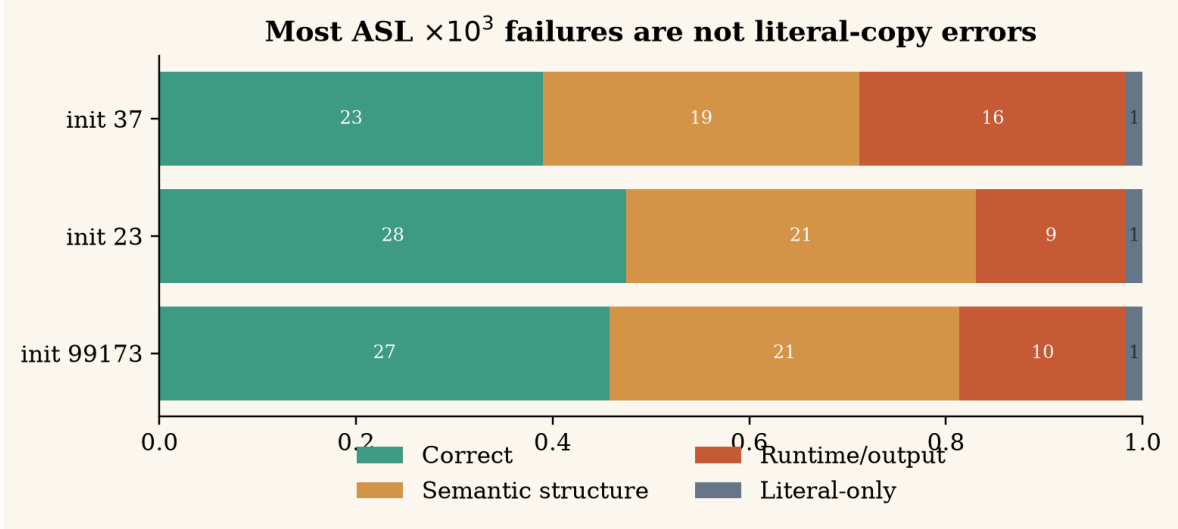


Figure 7: Mutually exclusive outcomes for the 59 factor-1,000 ASL predictions. Counts are printed within each seed’s bar. Literal-only oracle correction rescues one case per seed; executable semantic mismatch and runtime/output failure account for almost all remaining errors.

medium-, and 25 low-step cases). The larger 250/59 experiment remains the primary causal comparison; the 55-parent view is only a common-denominator curve. Before running the three new scales, provenance-checked projection of the identical cached $\times 1$ and $\times 10^3$ questions gives 36/55 to 23/55 for B1. The ASL seeds change from 27/55 to 25/55, 25/55 to 26/55, and 24/55 to 21/55, respectively. Thus the endpoint-only curve reproduces the large direct drop and comparatively stable ASL behavior, but remains interim until B1L and all five factors complete.

We also partition each of the 59 factor-1,000 ASL outcomes into correct, runtime/output failure, conservative literal-copy error, or remaining executable semantic mismatch. A literal-copy label requires a correct original prediction, matching literal-free alpha structure, and a correction that changes numeric literals only according to the frozen source/trace mapping. Initializations 99173, 23, and 37 yield, respectively, (27, 10, 1, 21), (28, 9, 1, 21), and (23, 16, 1, 19) cases in the order just listed. Oracle literal correction therefore adds only one answer per seed. This lower-bound diagnostic does not support numeric transcription as the main large-number failure source; semantic and runtime/output failures dominate.

The ASL arms of the frozen model-size gate are also complete. With the same U2000/E4500 data, rank-8 QKVO placement, and initialization 99173, Qwen3-1.7B improves the full ordinary score from 110/250 (44.0%) to 139/250 (55.6%). On the 59 paired parents it improves ordinary

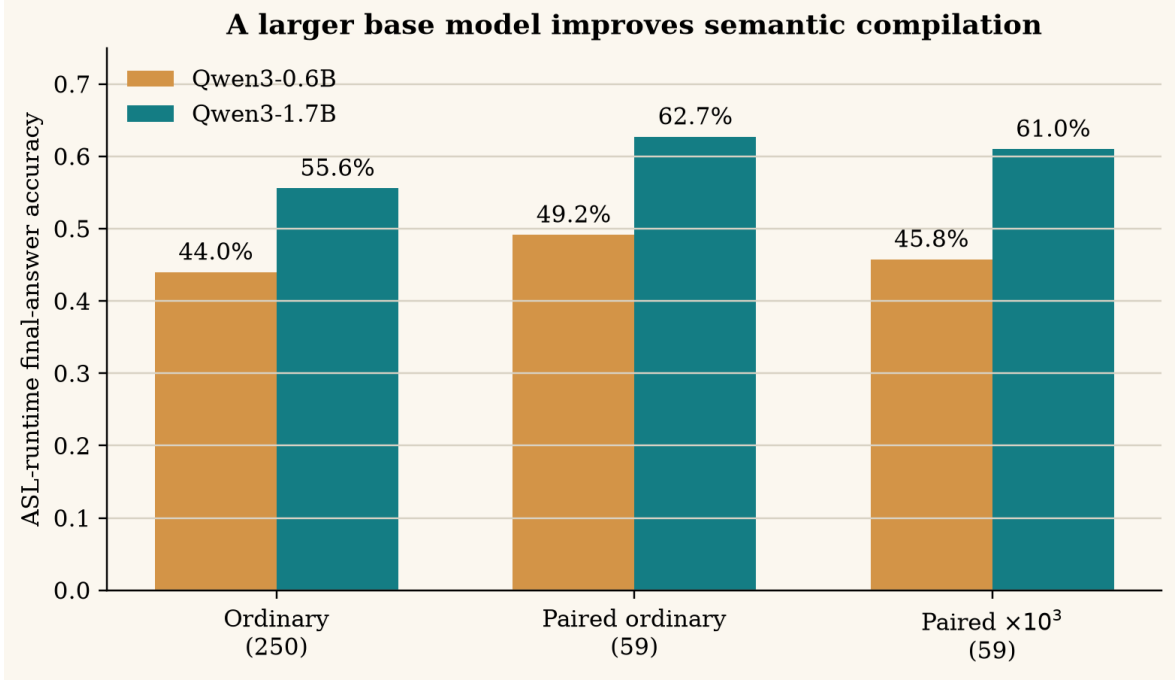


Figure 8: ASL-runtime scaling at matched training data, adapter placement, and initialization. Qwen3-1.7B improves over Qwen3-0.6B on the complete 250-question view and both endpoints of the fixed 59-parent robustness comparison.

accuracy from 29/59 (49.2%) to 37/59 (62.7%) and factor-1,000 accuracy from 27/59 (45.8%) to 36/59 (61.0%); the larger ASL model therefore loses only 1.7 points under the transformation. Figure 8 visualizes this consistent 11.6–15.3-point gain. The matched 1.7B direct arm remains in progress, so these completed ASL arms establish a semantic-compilation scaling signal but not yet an ASL-over-direct accuracy claim at 1.7B.

13.6 GSM8K-Only Diversity and Repetition

We train three G1-GSM8K adapters on the 4,500-unique-program freeze and evaluate only TEST-GSM17. All use the pinned Qwen3-0.6B revision, QKVO rank 8, 4,500 row exposures, 570 optimizer steps, and the same autonomous prompt and decoding seed. The first run uses initialization seed 99,173 on XPU; initialization seeds 23 and 37 run on CUDA. All use dataset-selection seed 11. Historical artifact paths retain the shorthand `seed11` for the first run, but that shorthand must not be interpreted as model-training seed 11. Backend provenance is retained per run. We additionally run an exposure-matched scale cell, U2000/E4500, which selects 2,000 unique GSM8K programs and reuses each two or three times in deterministic shuffled cycles. It uses dataset-selection seed 11 and is directly paired with the corresponding initialization-99,173 4,500-unique condition. Before observing official-test outcomes, we also freeze U1000/E4500: 1,000 unique source IDs and semantic signatures, each reused four or five times, with the same 4,500 exposures, ten logical epochs, and 570 optimizer steps. This predeclared cell isolates stronger repetition rather than responding post hoc to confirmation performance.

The 4,500-unique GSM8K condition obtains 6, 7, and 6 answers across initializations (mean 6.33/17, 37.3%; range 35.3–41.2%), improving over Q0’s 4/17 in every seed. Executable counts are 13, 13, and 14 versus 12, and mean alpha-state F1 values are .527, .559, and .556 versus .502.

Table 24: GSM8K-only scaling on the frozen 17-item historical projection. All trained conditions use 4,500 row exposures and 570 optimizer steps. U2000 is a three-initialization matched replication.

Condition	Parse	Lower	Type	Execute	α -return	Answer
Q0 Codex-450 repeated	14	13	13	12	4	4
G1 4,500 unique, init 99173	16	16	15	13	5	6
G1 4,500 unique, init 23	16	16	16	13	6	7
G1 4,500 unique, init 37	16	16	16	14	5	6
U2000, init 99173	16	16	16	15	6	11
U2000, init 23	15	15	15	14	5	6
U2000, init 37	15	15	15	14	6	6

However, these GSM8K-only scores are below the historical mixed-D1 GSM8K counts of 10, 8, and 11. This comparison does not isolate a beneficial effect of TAT-QA: removing TAT-QA also changes the selected GSM8K identities and their exposure schedule.

The exposure-matched U2000 answer counts are 11, 6, and 6 (mean 7.67/17, 45.1%; range 35.3–64.7%), compared with same-initialization G1-4500 counts of 6, 7, and 6. Thus the matched answer deltas are +5, −1, 0: initialization 99,173 has five gains and no losses (two-sided exact McNemar $p = .0625$, exploratory), initialization 23 has two gains and three losses ($p = 1.0$), and initialization 37 has two gains and two losses ($p = 1.0$). The large first-run gain therefore does not replicate uniformly.

U2000 nevertheless remains above Q0’s 4/17 in every seed. Its executable counts are 15, 14, and 14, and mean alpha-state F1 values are .620, .571, and .572, versus G1’s 13, 13, and 14 executable programs and .527, .559, and .556 F1. The more consistent effect is therefore soft alpha-normalized semantic overlap, not final answers. These data reject both a monotonic “more unique is always better” rule and a confirmed U2000 optimum. The predeclared U1000 cell and the larger official confirmation are required to distinguish a real diversity–repetition tradeoff from high seed and small-test variance.

13.7 Greedy Semantic Augmentation

We next apply the frozen development-only augmentation gate to the initialization-99,173 U2000 checkpoint. The 250-case selection view is disjoint from final confirmation and remains fixed across increments. AUG1 adds execution-preserving relation paraphrases; it ties the parent at 99/250 (39.6%) and is rejected. AUG2 therefore warm-starts from U2000 rather than from AUG1 and adds synchronized entity renaming. It reaches 101/250 (40.4%), with 21 candidate-only and 19 parent-only successes, and passes the prespecified strict-improvement rule by two answers. The paired difference is small and not statistically resolved (two-sided exact McNemar $p = .875$), so this is an exploratory carry-forward decision rather than evidence of a general gain.

The separate 17-case historical diagnostic moves in the opposite direction: U2000 obtains 11/17 whereas AUG2 obtains 7/17, with four parent-only and no candidate-only successes. Candidate alpha-state F1 also falls from .620 to .539. This disagreement is retained rather than optimized away. It indicates that entity renaming may modestly help the broader frozen selection sample while damaging some previously successful structures, and motivates replication at both model sizes before making a semantic-augmentation claim.

Table 25: Matched O1 operator pilot. Entries are correct/count; tokens are mean generated tokens. The generic-tool arm remains pending.

Condition	Overall	Square	Square root	Cube	Tokens
Direct reasoning	86/100	30/33	33/33	23/34	464.0
Primitive ASL (AP)	100/100	33/33	33/33	34/34	14.5
Semantic ASL (AS)	66/100	33/33	33/33	0/34	16.5

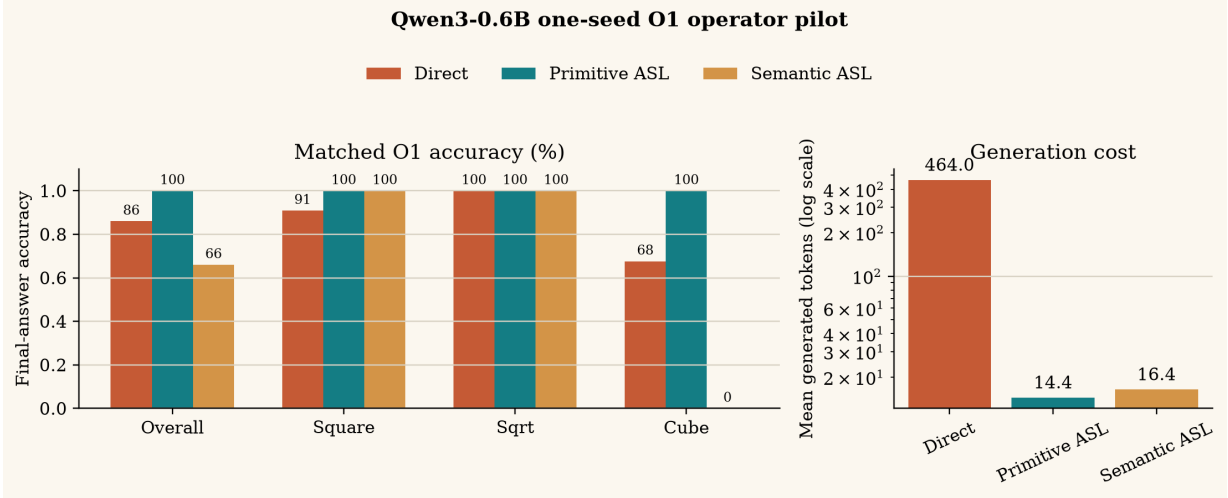


Figure 9: O1 answer accuracy by operator family and generation cost. Token cost uses a logarithmic axis. AP is perfect in this pilot; AS exposes a systematic cube-argument binding failure.

13.8 One-Seed O1 Operator Pilot

We next freeze a pilot designed to obtain an early read before running the full operator-complexity ladder. Qwen3-0.6B sees 250 training and 30 development records for three exact unary families: square, cube, and perfect square root. The 100 test questions use disjoint surface-template families and contain 33, 34, and 33 cases, respectively. AP denotes a primitive ASL vocabulary (square, cube, and sqrt); AS denotes semantic aliases (area_of_square, volume_of_cube, and side_of_square). Each rank-8 QKVO adapter receives 750 total exposures. Direct reasoning uses the same base revision, identities, inference seed, and hidden answers, with native thinking enabled. This is a synthetic, single-seed behavior pilot rather than confirmatory evidence.

AP gains 14 paired answers and loses none relative to direct reasoning (two-sided exact McNemar $p = 0.00012$, exploratory). Its mean output is 32.1 times shorter, because the model selects and binds an operator while the runtime performs exact execution. The direct deficit is concentrated in cube (67.6%) and, to a lesser degree, square (90.9%); exact roots are perfect. Thus the result is consistent with operator delegation helping as direct calculation becomes less reliable, but O0 and harder matched families are required to measure a complexity interaction.

AS provides the corresponding warning. All outputs parse, lower, and type-check, but all 34 cube outputs create `side` and invoke `volume_of_cube(edge)`. The undefined `edge` prevents execution, while every square and root case succeeds. The 34 errors are therefore a single systematic argument-binding failure, not evidence that semantic operators are intrinsically worse. The frozen result is retained unchanged; a separately labeled symbol-grounding diagnostic will test the repair without overwriting the pilot.

Table 26: Exploratory XPU pilot. Accuracy intervals are 95% Wilson intervals; tokens and latency are means. Latency reflects the correctness-first full-prefix adapter and is not a production estimate.

Condition	Accuracy	Trigger recall	Output tokens	Latency (ms)
LLM only	0.750 [0.409, 0.929]	—	40.56	8028
Matched prompt	0.625 [0.306, 0.863]	—	45.81	9380
Explicit tool	0.875 [0.529, 0.978]	0.625	22.50	4410
Reflex	0.750 [0.409, 0.929]	0.750	46.06	9195
Oracle	1.000 [0.676, 1.000]	1.000	15.31	2718

Natural-parent operator correction. The preceding pilot isolates operator invocation and runtime execution, but its single-operation templates do not measure the multi-entity binding and dependency errors that dominate natural GSM8K compilation. We therefore retain it only as a synthetic sanity baseline. The primary operator experiment is now frozen as **GSM8K-0C-natural-v2**: 250 training, 30 development, and 100 test parents are drawn from execution-verified GSM8K programs excluded from all 2,000 U2000 training identities. The eligibility audit found 766 held-out parents and selected the splits before inference.

For each parent, one grounded source binding is shared across all interventions. The remainder of the original question and downstream ASL dependency graph is preserved. The panel includes the original problem; strict multipliers $\{1, 10^2, 10^3, 10^4, 10^6\}$; three deterministic Uniform $[-0.30, +0.30]$ jitter seeds; and value-preserving O1, O3, O5, and O6 subgraphs using exact square root, base-10 logarithm, linear solution, and positive quadratic-root operators. Every transformed answer is recomputed by executing the full transformed program. Direct and ASL conditions must consume the same frozen row and match both ordered identity and question hash. Results from this panel supersede the synthetic pilot for the principal direct-versus-ASL crossover analysis once both arms complete.

14 Preliminary XPU Pilot

14.1 Setup

We ran a diagnostic pilot with Qwen3-0.6B at pinned revision `c1899de` (the full hash is stored in the manifest), PyTorch 2.13.0+XPU, and an Intel integrated GPU. The deterministic sample crossed one and three operators with one- and two-digit operands, using addition and multiplication. It contained two examples per cell (eight arithmetic examples) and eight matched non-trigger controls. One generation seed (17) and five conditions yielded 80 generations. The 96-token output budget was selected after a shorter plumbing run exposed truncation; the shorter run is not included in the results. Confidence intervals are Wilson intervals over only eight arithmetic observations per condition and are therefore intentionally wide.

14.2 Results

The reflex condition did not improve exact-match accuracy over LLM only in this pilot. Explicit-tool accuracy was numerically higher and oracle intervention was perfect, but the sample is too small for either difference to support a general claim. No condition falsely intervened on a control. Accepted explicit-tool candidates represented the intended expression in every triggered case, whereas only 50% of reflex candidates did. Inspection showed that the strict watcher often intercepted a valid intermediate subexpression rather than the full task expression. Once a candidate

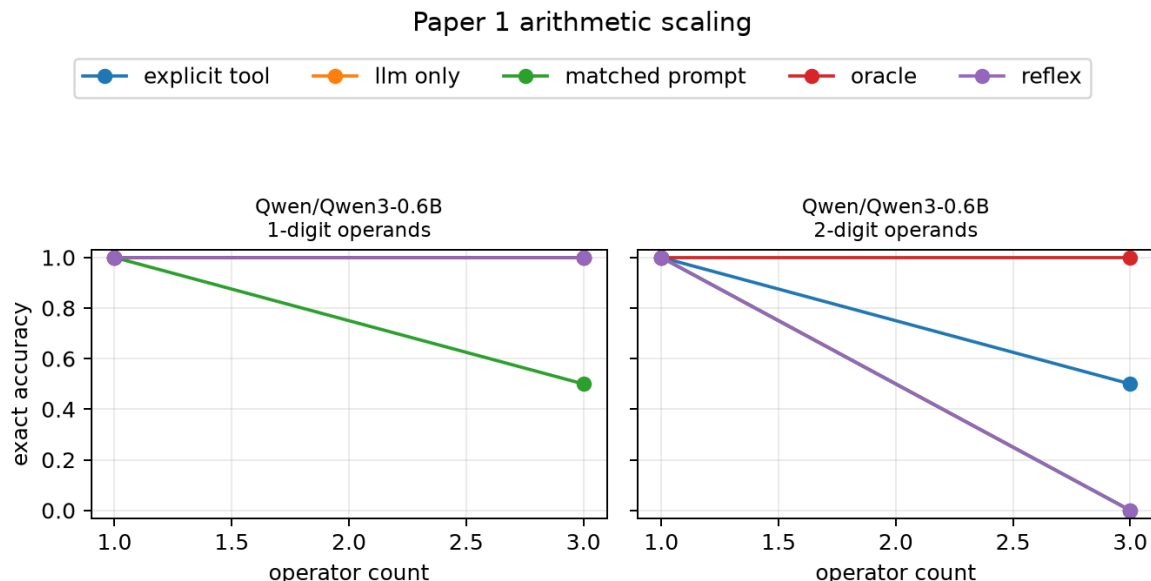


Figure 10: Exploratory exact-match accuracy by generated difficulty cell. Each point contains only two examples; the figure diagnoses plumbing and candidate behavior and must not be read as a scaling estimate.

reached the engine, execution correctness was 100% in both conditions. Reflex result-use was 66.7%, and one third of used results were later overridden. These component results locate the present bottleneck in exposure, selection, and downstream use rather than arithmetic execution.

At this protocol revision H1 is not supported: the automatic reflex tied the unassisted baseline and failed to dominate explicit invocation. H2–H4 remain untested because this pilot includes neither a production interface nor the required model-size and seed sweeps. Paper 2 therefore remains gated. A confirmatory run must increase examples per cell, use all declared seeds and model sizes, add a second family, and freeze any detector change before examining outcomes.

15 Falsification and Evidence Gate

The reflex claim is not supported if any of the following holds under matched capability and declared cost accounting:

- answer gains disappear when final model claims, rather than inserted engine values, are scored;
- false triggers, normalization errors, or ignored results offset exact execution gains;
- explicit calculator calling matches or dominates the quality–token–latency frontier;
- reflex and baseline accuracy have indistinguishable difficulty slopes;
- gains occur only with oracle detection or only in one easy/model cell; or
- results fail to reproduce across seeds or a second model family.

Paper 2 is justified only if Paper 1 identifies a reproducible regime where automatic strict assistance helps and mixed operation types expose a limitation of one calculator. Paper 3 is

not justified by syntax brittleness alone; learned interrupts require a measured task regime worth extending.

16 Reproduction

The reusable runtime lives under `src/ccpu/common`; Paper 1 code lives under `src/ccpu/paper1`. From the repository root, the dependency-free protocol check is:

```
python -m pip install -e .
python -m pytest
python -m ccpu paper1 generate \
    --config configs/paper1/smoke.json \
    --output artifacts/paper1/smoke/dataset.jsonl
python -m ccpu paper1 validate \
    --dataset artifacts/paper1/smoke/dataset.jsonl
python -m ccpu paper1 simulate \
    --dataset artifacts/paper1/smoke/dataset.jsonl \
    --output-dir artifacts/paper1/smoke/run
```

The pinned pretrained path is:

```
python -m pip install -e ".[hf]"
python -m ccpu paper1 generate \
    --config configs/paper1/primary.json \
    --output artifacts/paper1/primary/dataset.jsonl
python -m ccpu paper1 run-hf \
    --dataset artifacts/paper1/primary/dataset.jsonl \
    --config configs/paper1/primary.json \
    --output-dir artifacts/paper1/primary/run
```

The reported XPU pilot uses the direct interpreter from the validated XPU environment and can be reproduced with:

```
$py = 'C:\Users\j.simao\.venvs\modal-llm-xpu\Scripts\python.exe'
$env:PYTHONPATH = (Resolve-Path src).Path
& $py -m ccpu paper1 generate \
    --config configs/paper1/xpu_pilot.json \
    --output artifacts/paper1/xpu_pilot/dataset.jsonl
& $py -m ccpu paper1 run-hf \
    --dataset artifacts/paper1/xpu_pilot/dataset.jsonl \
    --config configs/paper1/xpu_pilot.json \
    --output-dir artifacts/paper1/xpu_pilot/run96
& $py -m ccpu paper1 replay \
    --dataset artifacts/paper1/xpu_pilot/dataset.jsonl \
    --completions artifacts/paper1/xpu_pilot/run96/predictions.jsonl \
    --output-dir artifacts/paper1/xpu_pilot/final
```

The held-out comparison and derived analysis use:

```

& $py -m ccpu paper1 run-hf \
  --dataset artifacts/paper1/hard_heldout_xpu/dataset.jsonl \
  --config configs/paper1/hard_heldout_xpu.json \
  --output-dir artifacts/paper1/hard_heldout_xpu/run \
  --device xpu
& $py -m ccpu paper1 paired \
  --dataset artifacts/paper1/hard_heldout_xpu/dataset.jsonl \
  --predictions artifacts/paper1/hard_heldout_xpu/run/predictions.jsonl \
  --output artifacts/paper1/hard_heldout_xpu/run/paired_analysis.json
python -m ccpu paper1 plot-interfaces \
  --summary artifacts/paper1/hard_heldout_xpu/run/summary.json \
  --output docs/papers/paper1/paper1_heldout_interfaces.png

```

Device selection for `device`: `auto` is CUDA, then XPU, then CPU. The replay step applies the current frozen evaluator to preserved generations and records both the source prediction hash and rescore provenance.

Use `--limit`, `--model`, and repeated `--condition` arguments for resource preflight. Each run writes predictions, full traces, summary tables, hashes, configuration, model metadata, environment information, and git revision/dirty status. The replay command permits model generation in a separate harness while preserving identical controller and evaluation semantics. The `plot` command renders the machine-readable scaling cells as accuracy curves by model, operand width, condition, and operator count. The `compare-models` command consumes the frozen comparison configuration and regenerates the cross-model table plus block-capability, interface-accuracy, token-latency, and failure-mode figures without manually entered results. The `generate-lora-data` and `train-lora` commands implement the versioned protocol-only adapter phase; the former must pass its leakage audit before the latter can run. The `analyze-placement` command hashes the source summaries and training reports and regenerates Table 7 and Figure 5 inputs.

The public transfer run is resumable by condition and is reproduced with:

```

& $py -m ccpu paper1 freeze-public-gsm8k \
  --source-selection artifacts/paper2/public_benchmarks/data_v1/selection.jsonl \
  --output-dir artifacts/paper1/public_gsm8k_v1/data
foreach ($condition in @('llm_only','matched_icl','generic_compute', \
  'calculator_block','runtime_trigger','oracle_calculator', \
  'lora_calculator_block')) {
  & $py -m ccpu paper1 run-public-gsm8k \
    --public-config configs/common/public_benchmarks.json \
    --cache-root C:\Users\j.simao\.cache\ccpu-public-v1 \
    --selection artifacts/paper1/public_gsm8k_v1/data/selection.jsonl \
    --model-config configs/paper1/public_gsm8k_qwen_xpu.json \
    --condition $condition --checkpoint-every 10 \
    --output-dir artifacts/paper1/public_gsm8k_v1/qwen3_0_6b/$condition
}
$analysis = @('paper1','analyze-public-gsm8k','--output-dir', \
  'artifacts/paper1/public_gsm8k_v1/analysis')
foreach ($condition in @('llm_only','matched_icl','generic_compute', \
  'calculator_block','runtime_trigger','oracle_calculator', \
  'lora_calculator_block')) {

```

```

$analysis += @('--predictions', \
    "artifacts/paper1/public_gsm8k_v1/qwen3_0_6b/$condition/predictions.jsonl")
}
& $py -m ccpu @analysis

```

The ASL dataset bootstrap uses verified local public-source files:

```

DSL=artifacts/paper1/dsl
python -m ccpu.dsl_dataset mine \
    --source gsm8k=<pinned-gsm8k-train.parquet> \
    --source tatqa=<pinned-tatqa-development.json> \
    --source clutrr=<pinned-clutrr.parquet> \
    --source proofwriter=<pinned-proofwriter.parquet> \
    --source-split tatqa=development \
    --source-split clutrr=test \
    --source-split proofwriter=test \
    --output-dir artifacts/paper1/dsl/raw_v1
python -m ccpu.dsl_dataset audit-chops \
    --input-dir artifacts/paper1/dsl/raw_v1 \
    --output artifacts/paper1/dsl/raw_v1/chop_audit.json
python -m ccpu.dsl_dataset select \
    --input artifacts/paper1/dsl/raw_v1/gsm8k.jsonl \
    --max-examples 100 \
    --output artifacts/paper1/dsl/bootstrap_v1/gsm8k_seed.jsonl
python -m ccpu.dsl_dataset bootstrap-annotated \
    --input artifacts/paper1/dsl/bootstrap_v1/gsm8k_seed.jsonl \
    --input artifacts/paper1/dsl/bootstrap_v1/tatqa_seed.jsonl \
    --output-dir artifacts/paper1/dsl/operation_ledger_v1
python -m ccpu.dsl_dataset prepare-local \
    --seed artifacts/paper1/dsl/bootstrap_v1/gsm8k_seed.jsonl \
    --seed artifacts/paper1/dsl/bootstrap_v1/tatqa_seed.jsonl \
    --output-dir artifacts/paper1/dsl/local_codex_v1
python -m ccpu.dsl_dataset validate-semantic \
    --seed artifacts/paper1/dsl/bootstrap_v1/gsm8k_seed.jsonl \
    --seed artifacts/paper1/dsl/bootstrap_v1/tatqa_seed.jsonl \
    --annotations ${DSL}/local_codex_v1/annotations_primary.jsonl \
    --annotations ${DSL}/local_codex_repair_v1/annotations_repair1.jsonl \
    --annotations ${DSL}/local_codex_repair_v2/annotations_repair2.jsonl \
    --output-dir artifacts/paper1/dsl/annotated_v1
python -m ccpu.dsl_dataset generate --dry-run \
    --input artifacts/paper1/dsl/bootstrap_v1/gsm8k_seed.jsonl \
    --config configs/paper1/dsl_teacher.example.json \
    --skill skills/ccir-arith-compiler/SKILL.md \
    --output-dir artifacts/paper1/dsl/teacher_dry_run

# Relation-diversified 500-original checkpoint
python -m ccpu.dsl_dataset select-diverse \
    --input ${DSL}/raw_v1/gsm8k.jsonl \

```

```

--input ${DSL}/raw_v1/tatqa.jsonl \
--exclude ${DSL}/annotated_v1/accepted.jsonl \
--dataset-target gsm8k=267 --dataset-target tatqa=133 \
--output ${DSL}/expansion500_v1/candidates.jsonl
python -m ccpu.dsl_dataset prepare-local \
--seed ${DSL}/expansion500_v1/candidates.jsonl \
--output-dir ${DSL}/expansion500_v1/primary
python -m ccpu.dsl_dataset run-local \
--requests-dir ${DSL}/expansion500_v1/primary/requests \
--output-dir ${DSL}/expansion500_v1/primary \
--prompt configs/paper1/local_codex_annotation_prompt.md \
--schema configs/paper1/semantic_annotation_batch.schema.json
python -m ccpu.dsl_dataset validate-semantic \
--seed ${DSL}/expansion500_v1/candidates.jsonl \
--annotations ${DSL}/expansion500_v1/primary/annotations.jsonl \
--output-dir ${DSL}/expansion500_v1/validated_primary
python -m ccpu.dsl_dataset prepare-repair \
--seed ${DSL}/expansion500_v1/candidates.jsonl \
--rejected ${DSL}/expansion500_v1/validated_primary/rejected.jsonl \
--output-dir ${DSL}/expansion500_v1/repair1 --repair-round 1
python -m ccpu.dsl_dataset run-local \
--requests-dir ${DSL}/expansion500_v1/repair1/requests \
--output-dir ${DSL}/expansion500_v1/repair1 \
--prompt configs/paper1/local_codex_repair_prompt.md \
--schema configs/paper1/semantic_annotation_batch.schema.json
python -m ccpu.dsl_dataset validate-semantic \
--seed ${DSL}/expansion500_v1/candidates.jsonl \
--annotations ${DSL}/expansion500_v1/primary/annotations.jsonl \
--annotations ${DSL}/expansion500_v1/repair1/annotations.jsonl \
--output-dir ${DSL}/expansion500_v1/validated_repair1
python -m ccpu.dsl_dataset finalize-expansion \
--accepted ${DSL}/expansion500_v1/validated_repair1/accepted.jsonl \
--existing ${DSL}/annotated_v1/accepted.jsonl \
--frozen-ledger artifacts/paper1/asl_pilot_v1/freeze/split_ledger.jsonl \
--output-dir ${DSL}/expansion500_v1/freeze --target 350

```

17 Limitations

The elicitation instruction asks reflex models to expose an expression followed by an equals sign. This is less explicit than an API call but still changes generation behavior, and a model that answers directly may never trigger. Both trigger rate and conditional-on-trigger accuracy are therefore necessary. The strict lexical grammar does not understand intent and will miss prose arithmetic; expanding semantics belongs to Paper 3. Quote suppression is deliberately limited, and code-like or unusual formatting remains an adversarial surface.

The synthetic task controls arithmetic complexity but not realistic discourse. The textual explicit-tool baseline may understate a model’s native structured function-calling ability. Full-prefix token recomputation overstates runtime cost. The main comparisons have one greedy seed,

16 arithmetic examples, and four observations per selected cell. Their intervals remain wide, the adaptive bands are coarse, and no architecture or universal scaling generalization is supported. The 4B row is only a six-item smoke. Llama remains unavailable behind manual authorization; approved Gemma3-1B and ungated SmoLLM2 provide two cross-family adapter replications but cannot establish broad family effects. The adapter data are synthetic and small, and near-zero development loss may reflect the narrow protocol distribution. Twelve controls do not tightly bound rare false triggers, as the ICL failures illustrate. Adapter+minimal was selected by the predeclared gate, but the three-family comparison remains developmental rather than confirmatory. The zero-shot block prompt’s historical result is not reinterpreted by the separately developed ICL prompt, endpoint audit, or LoRA result. Finally, exact calculator output cannot repair an incorrectly copied expression, a model that ignores the result, or a task whose premises are wrong. The public transfer slice adds further limits: it has one small checkpoint, one greedy seed, and no no-compute controls. Operation-count strata are not a complete difficulty model. Source rationales provide noisy opportunity annotations, and 10.1% of annotated operations fall outside the registered calculator surface. The concrete syntax example causes strong anchoring in generic and automatic conditions; because it was frozen before the full outcome, this is valid evidence for that protocol but not evidence that all tool or trigger prompts must fail. The ASL corpus is suitable only for the explicitly labeled Q1 pilot. Although all 500 frozen semantic programs execute to the benchmark answer, 176 required rationale-assisted repair (52 in the initial checkpoint and 124 in the expansion), and deterministic lint cannot establish that every entity, quantity, or temporal field is the best semantic representation. No row is Q4 manual gold. The grouped split prevents measured topology leakage but does not replace independent-teacher agreement or manual semantic adjudication. Twenty-five untouched test programs also give imprecise whole-program estimates; the richer component metrics are diagnostic rather than extra independent trials. The 900 perturbations share only 100 semantic parents and LoRA-B has roughly twice LoRA-100’s target-token exposure, so robustness differences cannot be attributed to augmentation alone. The normalized pattern signature is useful for leakage control but may merge or separate structures imperfectly. The 500-original checkpoint adds 350 relation-diversified training programs while excluding every frozen development/test semantic-pattern family. Its QKVO-r8 result changes semantic coverage and total token exposure together. Incremental NL+executed-state training and the matched rank-16 ablation show that doubling QKVO rank does not move held-out answers. QKVO+MLP placement, context preservation, epoch selection, and base-model scale remain developmental tests.

18 Related Work

Toolformer trains language models to decide when and how to invoke APIs, including a calculator [1]. PAL delegates model-generated programs to an interpreter [2], while ReAct interleaves model-authored reasoning and actions [3]. These systems establish the value of external execution. Paper 1 isolates a different control point: a runtime recognizes ordinary strict arithmetic already emerging in the output and invokes a bounded engine without a function-name decision. The comparison to explicit invocation is empirical; the paper does not assume that the reflex interface is better.

LoRA freezes base weights and injects trainable low-rank matrices [4]; our use of it is conventional, and the empirical question is only whether a small adapter can store this interface contract. Self-RAG-like systems show that models can learn control or reflection tokens for inference actions [5]. Calculator fences are a typed local execution region, not a claim to have invented trained action syntax or control tokens.

19 Conclusion

Paper 1 reduces the cognitive-coprocessor proposal to its smallest credible experiment. A character-incremental detector, typed allowlisted micro-IR, exact bounded calculator, append-only state, and immediate text reinjection are enough to test whether automatic computation can improve an autoregressive stream. The original strict watcher does not improve accuracy, and the zero-shot fenced block usually emits invalid content. Deterministic surface normalization is numerically stronger, executing 12 of 16 full expressions and improving five paired examples without a loss, but remains an unreplicated result. Separate development shows that the block failure was not intrinsic to the grammar: one frozen two-shot prompt reaches 100% arithmetic execution on Qwen3-0.6B and 93.8% on Qwen3-1.7B. Capability changes the error type: 0.6B falsely blocks two controls, while 1.7B has no false block and one arithmetic miss. A clean 4B smoke supports feasibility but not scaling. Paper 1 therefore establishes a plausible few-shot semantic execution interface and a protocol-only adapter result across three small-model families. Adapter+minimal reaches perfect execution and observed control safety in all three; it reaches perfect answers in Qwen and SmolLM and 15/16 in Gemma, where residual ICL restores 16/16. Base+minimal never opens a valid arithmetic block in any family. Context enables cold-start behavior but costs more tokens and can produce false interventions; after adaptation it is redundant or harmful except for the measured Gemma residual. The normalized runtime is training-free but remains limited by model expression exposure and result integration. Gemma and SmolLM oracle runs further show that exact reinjection alone does not guarantee result use. The resulting architecture is hybrid: context for rapid development and measured residuals, weights for a stable semantic contract, and the runtime for typed parsing, bounds, exact computation, and enforced use. Powered confirmatory data and an authorized Llama run remain necessary before stronger placement or family claims. On public GSM8K, however, this narrow adapter and the training-free automatic interface do not transfer: high activation coexists with 6.5% or lower operation recall and at most 3.3% accuracy, while the bounded-calculator oracle reaches 75.8%. Where interface knowledge should live is therefore conditional: syntax can live in context or weights and exactness in the runtime, but natural-language decomposition is not solved by any of the three in this experiment. The ASL bootstrap turns that negative result into a testable next stage without claiming success: public records and scopes are mined, a compact extensible surface parses to typed state, and all 150 Q1 semantic programs execute exactly. A frozen grouped pilot now tests whether context, weights, or controlled augmentation produce a measurable compilation signal. Context alone teaches the surface, 50 genuinely different originals improve semantic compilation, and three residual examples complement the 100-row adapter. Controlled perturbations mainly improve invariance, not untouched semantic generalization. The result supports a hybrid placement—context for local specialization, weights for a learned ASL contract, and the runtime for lowering, types, and deterministic execution—but its untouched 25-program test and non-manual Q1 labels deliberately bound the conclusion. The relation-diversified 500-original adapter improves to 11/25, but clause-local executed-state compilation regresses to 7/25 and oracle state reaches only 6/25. Full-question local compilation recovers 11/25 but does not exceed whole-program synthesis. Valid deltas remain semantically weak, particularly on multi-clause GSM8K. Fine-grained replay localizes the strict whole-program deficit to attribute grounding (.144 F1), dependency preservation (.113 F1), relation direction (12/30), and RETURN grounding (7/25), despite .695 operator F1; 6/20 state-inequivalent programs still obtain the correct answer. Matched adapter-capacity, memory-placement, semantic-weighting, ranking, and graph-bottleneck controls do not exceed the 11/25 whole-program control. In contrast, three exposure-matched 4,500-program diversity runs reach 16/25, 13/25, and 17/25 answers, with 23/25, 24/25, and 22/25 executable programs. All three exceed Q0 on answers, execution, alpha-normalized return, and mean alpha-state F1. Strict

state and return equivalence remain below the control in all three. The replicated result therefore supports semantic training diversity as the only tested intervention that improves autonomous answer generation, not a stronger architecture claim or a solved canonical grounding problem. On the larger matched confirmation, direct reasoning decisively exceeds the three ASL adapters on ordinary questions (59.6% versus 34.0–44.0%). ASL therefore does not add ordinary answer accuracy at 0.6B. Under the paired factor-1,000 rewrite, however, direct reasoning loses 23.7 points while ASL changes by only -5.1 to $+3.4$ points, yielding an 18.6–27.1-point relative robustness gain with all three paired intervals above zero. This is evidence for bounded numerical magnitude robustness. The completed long-budget sensitivity preserves this result; the matched 1.7B gate determines whether it survives greater model capacity. The subsequent O1 behavior pilot supplies a complementary operator signal: primitive ASL obtains 100/100 answers versus direct reasoning’s 86/100, with 32.1-fold fewer generated tokens. This does not yet establish a monotonic operator-complexity advantage: it uses synthetic one-operation questions and one seed. Moreover, semantic ASL reaches only 66/100 because a systematic **side/edge** binding mismatch breaks every cube execution. The next evidence gate is therefore a matched categorical O0/O1/O3/O5/O6 ladder, followed by independently sampled uniform $[-30\%, +30\%]$ numeric jitter.

A Deterministic Semantic Diagnostic Taxonomy

The replay analyzer emits multi-label records; categories are not mutually exclusive. Path decomposition treats the first path component as the entity, non-qualifier semantic tokens as attributes, and separately records temporal, cardinality, status, and grouping qualifiers. CCIR assignments provide target, ordered sources, root and nested operators, constants, directed edges, and the RETURN expression. This supports separate tests for participants, direction, noncommutative argument order, literal magnitude, inter- versus intra-object relations, direct source-fact attachment, premature literal collapse, and transitive shortcuts. Aggregation and temporal supports are reported only when the reference contains the corresponding deterministic cues.

Evaluator v2 keeps those strict metrics and adds a weaker alpha view. It recursively inlines references to local assignments, removes the local target names, flattens and sorts commutative operations, and compares the resulting signatures of constants, operators, and external references. The return metric compares the complete inlined expression; the state metric compares multisets of all inlined assignment expressions. An unresolved external reference retains its semantic family, so alpha normalization does not erase external grounding. The metric is deterministic and answer-blind: equal numeric outputs with different expression graphs are not alpha equivalent. It measures computational equivalence under local renaming, not canonical ontology naming, and is therefore always reported beside rather than instead of strict path-sensitive equivalence and final execution.

Standalone alignment is one-to-one and requires compatible entities, lexical semantic overlap, and consistent graph roles; ambiguity or an unresolved local ontology enters the judge queue. Stateful replay instead requires exact reuse of already established paths and records accidental renames, duplicate slots, and unresolved references. The optional semantic judge is not run in the reported results. Coreference target accuracy requires mention annotations that the frozen corpus does not contain, and unit compatibility requires a dimensional type system; both are marked not measurable rather than inferred from generic parse or type success. Regression fixtures cover wrong entities and attributes, qualifier-only mismatches, allowed standalone versus forbidden stateful renames, relation and argument reversal, wrong constants, literal collapse, transitive shortcuts, each/total confusion, temporal direction, RETURN grounding, and multi-label failures without remote calls.

References

- [1] T. Schick, J. Dwivedi-Yu, R. Dessi, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. Toolformer: Language models can teach themselves to use tools. *NeurIPS*, 2023. <https://arxiv.org/abs/2302.04761>.
- [2] L. Gao, A. Madaan, S. Zhou, U. Alon, P. Liu, Y. Yang, J. Callan, and G. Neubig. PAL: Program-aided language models. *ICML*, 2023. <https://arxiv.org/abs/2211.10435>.
- [3] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing reasoning and acting in language models. *ICLR*, 2023. <https://arxiv.org/abs/2210.03629>.
- [4] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. *ICLR*, 2022. <https://arxiv.org/abs/2106.09685>.
- [5] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR*, 2024. <https://arxiv.org/abs/2310.11511>.
- [6] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman. Training verifiers to solve math word problems. *arXiv:2110.14168*, 2021. <https://arxiv.org/abs/2110.14168>.